



Name: _____

Class: **Circle one of the following:**

12Physics25A2 (RP) 12Physics25A3 (JM^cH) 12Physics25A6 (JM^cH)

Merewether High School

2025

Year 12 Physics Trial Examination

General Instructions

- Reading time - 5 minutes
- Working time – 3 hours
- Write using black pen.
- Draw diagrams using pencil.
- Calculators approved by NESA may be used.
- A data sheet, formula sheet and Periodic Table are provided with this paper.
- Use your sticker with your NESA student number.

Total Marks:

100

Section I - 20 marks (pages 3 - 13)

- Attempt Questions 1-20
- Allow about 35 minutes for this section.
- Use the multiple-choice answer sheet provided.
- This section will be computer-marked. Use a 2B pencil to indicate your choice. If you change your mind, you must completely erase your initial choice.

Section II - 80 marks (pages 14-39)

- This section has been split into two parts. Place a NESA sticker at the front of each section as indicated.
- Attempt Questions 21 - 35
- Allow about 2 hours 25 minutes for this section.
- Answer the questions in the space provided. These spaces provide guidance for the expected length of response.
- **Whilst completing the paper, it is the student's responsibility to allocate time that is commensurate with the marks allocated to each question.**
- Show all relevant working in questions involving calculations.
- **Additional writing space** is available at the end of each part in **Section II**. If you use this space, **clearly indicate that you have done so** and write the **number of the question** you are completing in the spare writing space.

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PART I – 20 marks

Attempt Questions 1-20

Allow about 35 minutes for this part

Use the multiple-choice answer sheet for Questions 1-20

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely using a 2B pencil.

Example:

Question 1

$2 + 4 = ?$ A. 2 B. 6 C. 8 D. 9

Q.1 ☐ A ☐ B ☒ C ☐ D

If you think you have made a mistake, **FULLY ERASE** your incorrect selection, ensuring no marks are left, and then fill in the new answer.

Q.1 ☐ A ☒ B ☐ C ☐ D

-
1. Which of the following evidence suggests protons and neutrons are not fundamental particles?
- A. Protons have a mass of 1.67×10^{-27} kg.
- B. Neutrons have a zero net charge.
- C. Neutrons do not have their own magnetic field.
- D. Protons and neutrons can undergo beta decay.
2. A car turning around a horizontal circular bend starts sliding out of control.

Which of the following statements is incorrect?

- A. Centripetal acceleration is no longer towards the centre of the curvature of the bend.
- B. There is no change to the car's centripetal acceleration.
- C. The coefficient of static friction between the tyres and the road has not changed.
- D. The car has exceeded the maximum threshold velocity for the curvature of the track.

3. An ideal transformer has an input voltage of 150 V and an output voltage of 600 V.

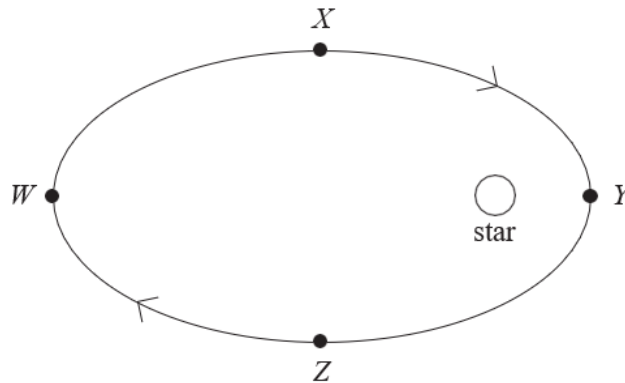
What is the ratio of turns in the primary coil to the number in the secondary coil?

- A. 1:4
 - B. 1:16
 - C. 4:1
 - D. 16:1
4. Which of the following statements best describes a feature of electromagnetic waves according to Maxwell's theory of Electromagnetism?
- A. An oscillating electric field creates a constant magnetic field parallel to the electric field.
 - B. An oscillating electric field creates an oscillating magnetic field, but an oscillating magnetic field does not create an electric field.
 - C. The found value of the speed of electromagnetic waves was different from the predicted value of the speed of light, so they were distinct phenomena.
 - D. An electromagnetic wave's electric field, magnetic field and direction of propagation were all perpendicular to one another.
5. Which of the following best represents the path of electricity from a power station to a home?
- A. Power station → step up transformer → high voltage transmission lines → step down transformer → low voltage transmission lines → home
 - B. Power station → step up transformer → low voltage transmission lines → step down transformer → high voltage transmission lines → home
 - C. Power station → step down transformer → high voltage transmission lines → step up transformer → low voltage transmission lines → home
 - D. Power station → step down transformer → low voltage transmission lines → step up transformer → high voltage transmission lines → home

6. An 8500 kg starship drifts towards a planet with a mass of 6.9×10^{27} kilograms and a radius of 72 000 km.

When the starship is 3000 km from the planet's surface, what is its acceleration, assuming the engines are turned off?

- A. 21.8 m s^{-2}
B. 40 m s^{-2}
C. 81.8 m s^{-2}
D. 162 m s^{-2}
7. A comet is in a stable elliptical orbit around a star, as shown in the diagram.

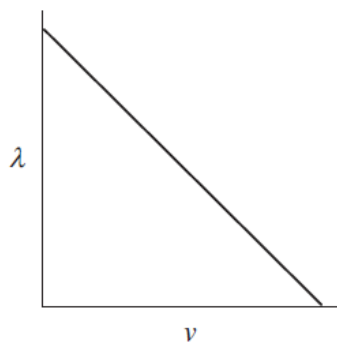


At which position does the comet have the greatest speed?

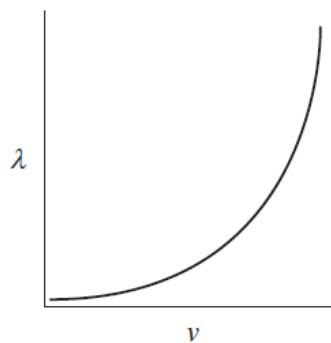
- A. W
B. X
C. Y
D. Z

8. Which graph represents the relationship between the de Broglie wavelength, λ , and speed, v ?

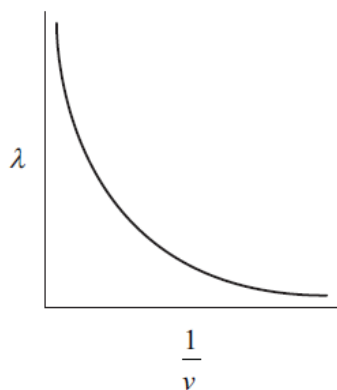
A.



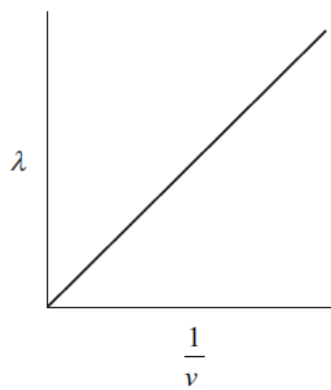
B.



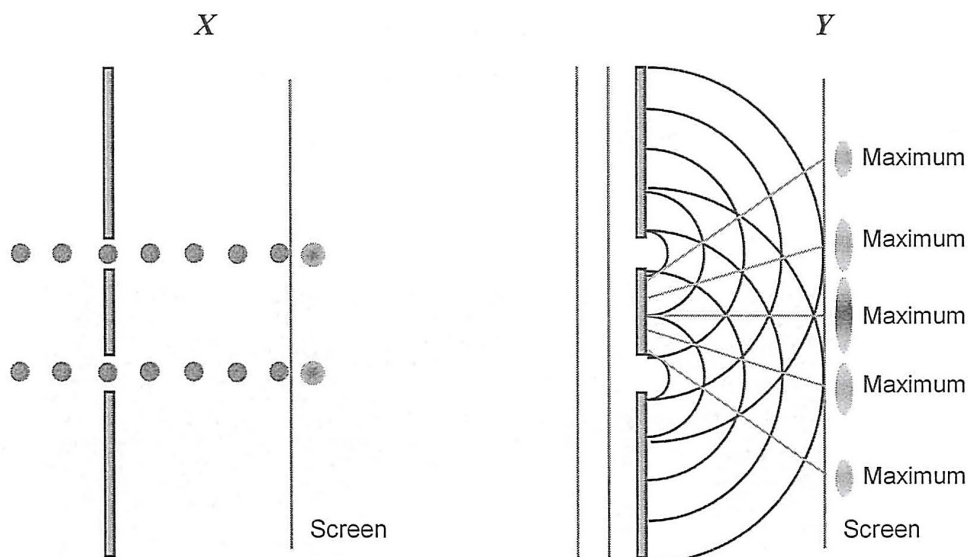
C.



D.



9. Diagrams *X* and *Y* below depict possible results of light entering two narrow slits which are very close together.

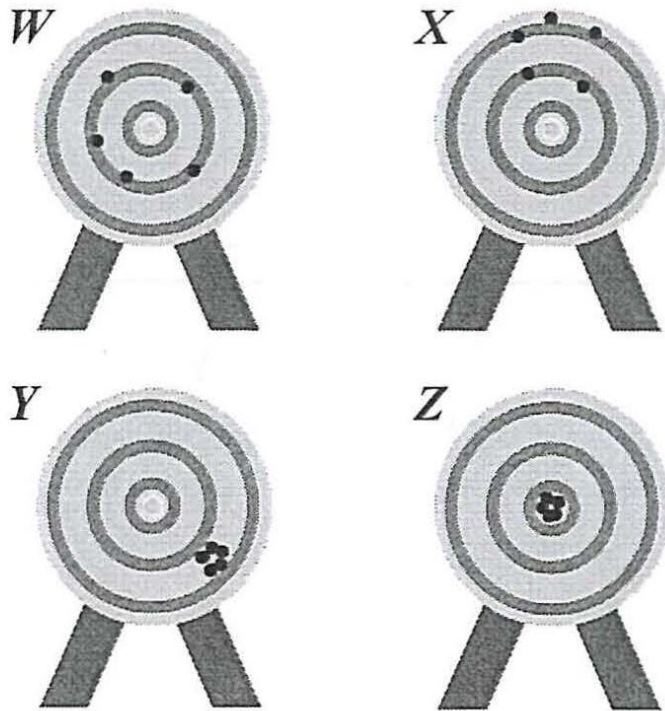


Which statement about these diagrams is correct?

- A. The particle model of light supports both diagrams.
- B. The wave model of light supports both diagrams.
- C. Diagram *X* shows the results of Young's double slit experiment.
- D. The particle nature of light is not supported by diagram *Y*.

10.

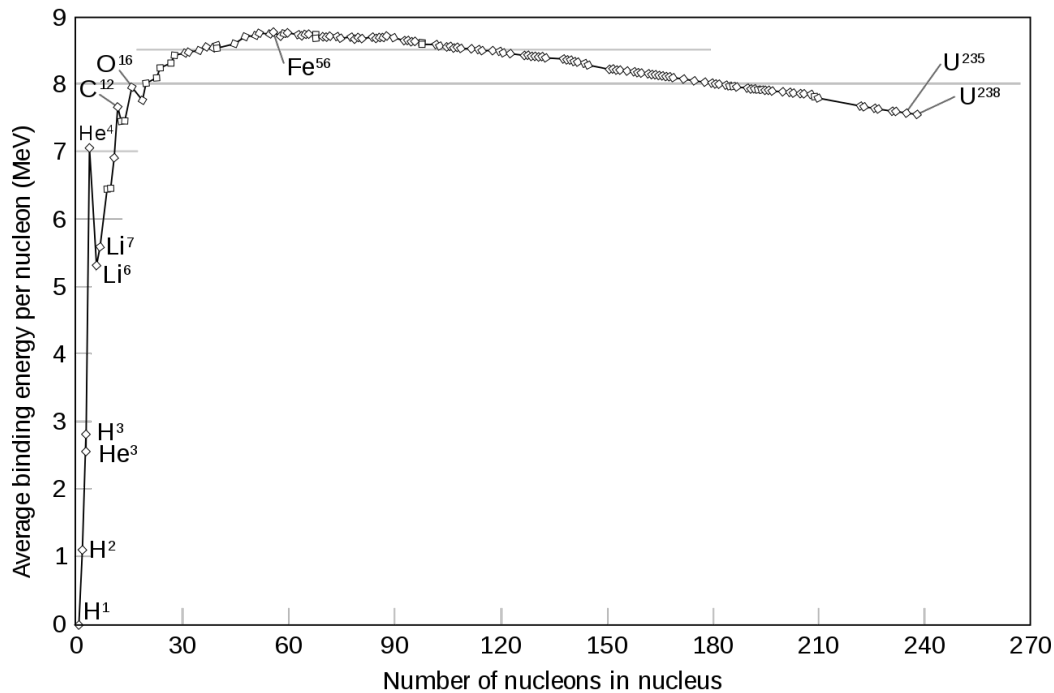
The diagrams show the results of darts being thrown at a board.



Which statement about these results is correct?

- A. Results W are more reliable than results Y.
- B. Results Y and Z are more reliable than W and X.
- C. Results Z are reliable, while results Y are unreliable.
- D. Results X are less reliable than results W.

11. The diagram shows the average binding energy per nucleon against the number of nucleons inside the nucleus.



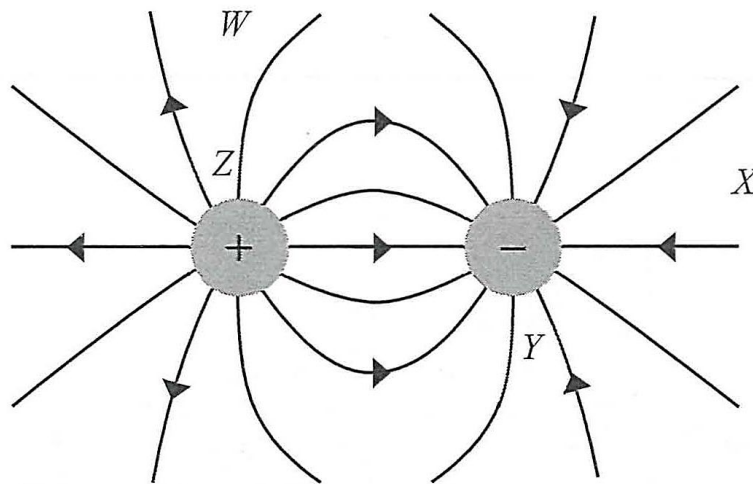
Source: https://commons.wikimedia.org/wiki/File:Binding_energy_curve_-_common_isotopes.svg

An exothermic reaction releases energy whereas an endothermic reaction absorbs energy from the environment.

Which of the following statements is correct? Had to change option D after the exam as had two answers the same

A.	Nuclear fusion resulting in He^4 is endothermic.	The decay of U^{238} into Th^{234} is endothermic.
B.	Nuclear fusion resulting in He^4 is exothermic.	The decay of U^{238} into Th^{234} is endothermic.
C.	The decay of U^{238} into Th^{234} is exothermic.	Nuclear fusion resulting in He^4 is exothermic.
D.	It cannot be determined whether fusion or fission reactions are endothermic or exothermic from the graphic given.	

12. Four positions are marked within the electric field shown.

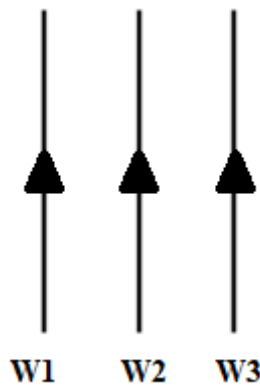


- A positive charge is moved from one of these positions to another position. Which move will involve the most work being done by the field?
- A. Moving the charge from *X* to *Y*
B. Moving the charge from *W* to *Z*
C. Moving the charge from *Y* to *W*
D. Moving the charge from *Y* to *Z*.
13. Two planets, *A* and *B*, orbit the same star. Planet *A*'s average orbital distance is six times larger than the average orbital distance of planet *B*. What is the relationship between the orbital periods, *T*, of the planets?
- A. $T_A = 6T_B$
B. $T_A = 15T_B$
C. $T_A = 36T_B$
D. $T_A = 216T_B$
14. What is the energy of a gamma ray photon produced from a single positron–electron annihilation?
- A. 0.5 eV
B. 1.0 eV
C. 0.5 MeV
D. 1.0 MeV

15. A circular coil of wire with a radius of 4.0 cm and 500 circular loops is placed in an electromagnet with the magnetic field decreasing at a rate of 0.200 T s^{-1} . The angle between the plane of the coil and the magnetic field is 30 degrees.

What is the magnitude of the induced emf? (A is the correct answer now but had to be changed after the exam from 0.213 V – was marked as B being correct as that was what had been entered into the marking system – saved changing the marking key)

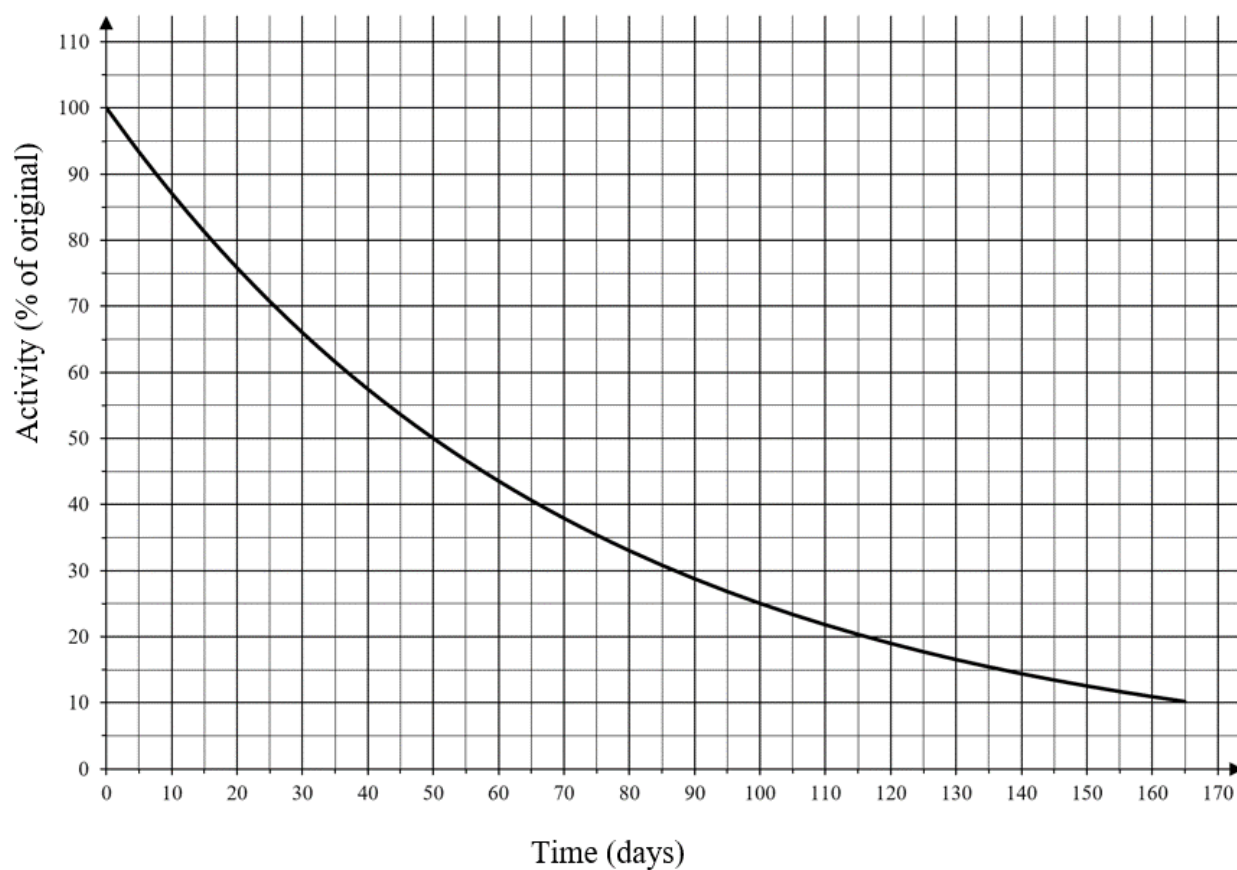
- A. 0.251 V
B. 0.435 V
C. 0.870 V
D. 1.50 V
16. Three parallel current carrying conductors result in the middle wire experiencing a net force of 0 N.



What would be the effect of doubling the current in W_3 AND changing the direction of the current in W_2 ?

- A. W_1 will repel W_2 and W_3 will repel W_2 , resulting in a non-zero net force on W_2 towards W_1 .
B. W_1 will repel W_2 and W_3 will repel W_2 , resulting in a zero net force on W_2 .
C. W_1 will repel W_2 and W_3 will attract W_2 , resulting in a non-zero net force on W_2 towards W_3 .
D. There will be no change to the system.

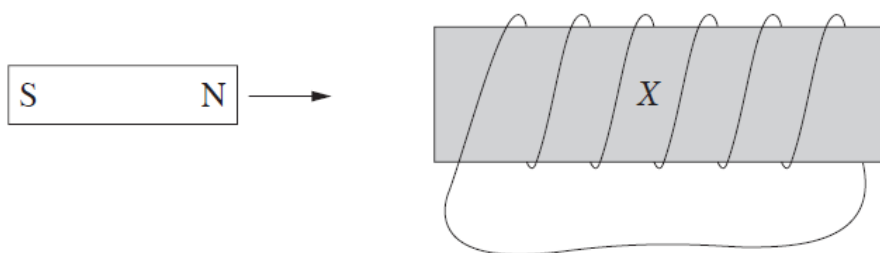
17. A radioactive sample's activity was observed to decay over time and the results plotted on the graph below.



The decay constant, λ , for this radioactive sample is closest to:

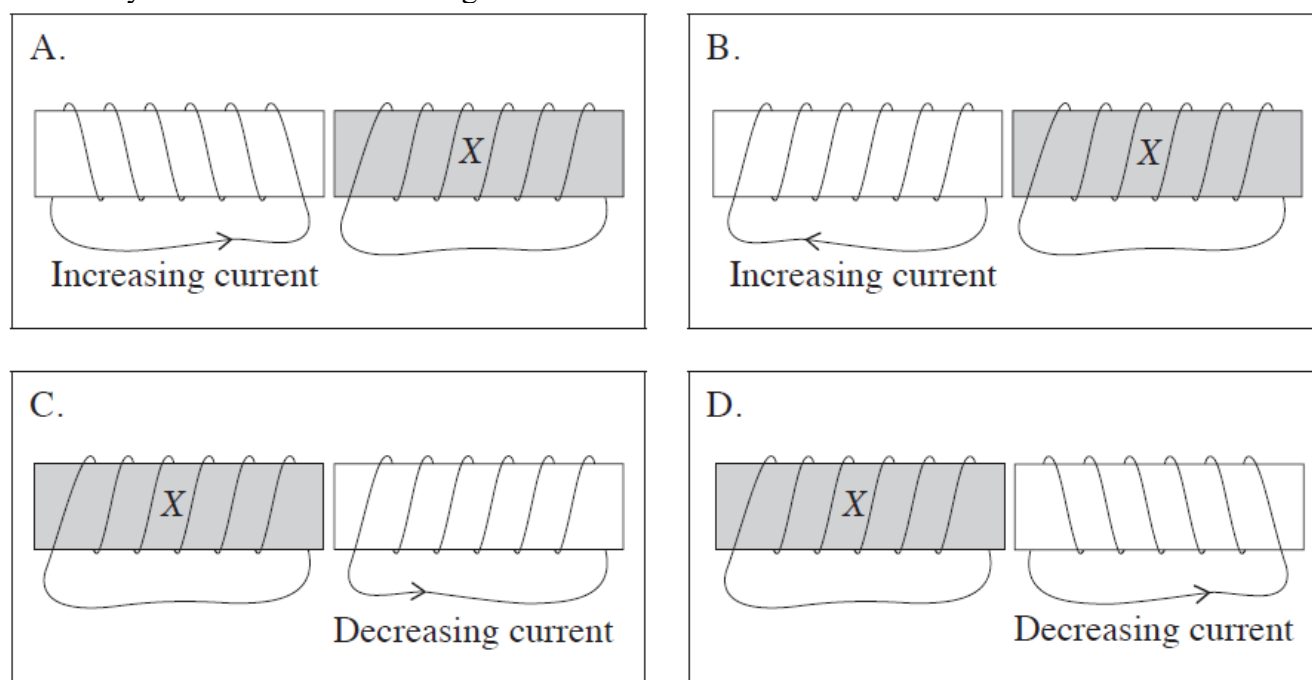
- A. $6.0 \times 10^{-3} \text{ s}^{-1}$
- B. $9.6 \times 10^{-6} \text{ s}^{-1}$
- C. $1.6 \times 10^{-7} \text{ s}^{-1}$
- D. $1.6 \times 10^{-7} \text{ day}^{-1}$

18. The diagram shows a magnet moving towards a coil X .



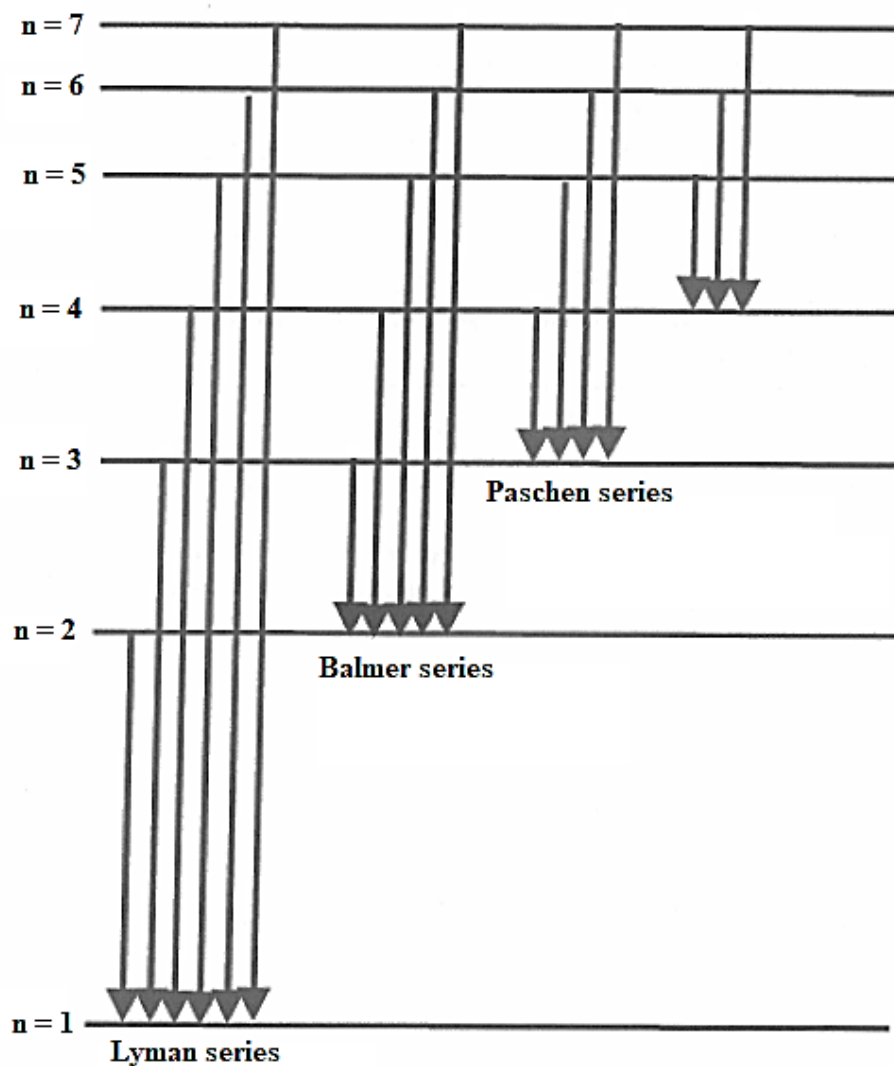
This action causes a current to be induced in the coil.

Which situation will induce a current in coil X that is in the same direction as the current induced by the movement of the magnet?



19. In a vacuum chamber there is a uniform electric field and a uniform magnetic field.
- A proton having a velocity, v , enters the chamber. Its velocity remains unchanged as it travels through the chamber.
- A second proton having a velocity, $2v$, in the same direction as the first proton, then enters the chamber at the same point as the first proton.
- In the chamber, the acceleration of the second proton
- is zero.
 - is constant in magnitude and direction.
 - changes in both magnitude and direction.
 - is constant in magnitude, but not direction.

20. The diagram below shows some of the transitions for three series of emission spectra for hydrogen.



An electron in the $n = 6$ energy level of a hydrogen atom relaxed to a lower energy level. In the process, a single photon with energy 3.025 eV was emitted.

Which of the following energy levels did the electron relax to?

- A. $n = 2$
- B. $n = 3$
- C. $n = 4$
- D. $n = 5$

END OF SECTION I



Student Number _____
AFFIX STICKER HERE IF AVAILABLE

Class: Circle one of the following

12Physics25A2 (RP) 12Physics25A3 (JM^cH) 12Physics25A6 (JM^cH)

Merewether High School

2025

Year 12 Physics Trial Examination

Section II Answer Booklet – Part A

Instructions

- Attach a sticker in the space above.
- This section has been split into TWO parts.
- Extra writing space is provided at the back of each part. If you use this space, clearly indicate which question you are answering.
- **When completing the paper, it is the student's responsibility to allocate time that is commensurate with the marks allocated to each question.**
- Write using a black pen.
- Draw diagrams using pencil.
- Calculators approved by NESA may be used.
- Show all relevant working in questions involving calculations.
- A data sheet, formulae sheets and Periodic Table are provided with this paper.
- Answer the questions in the space provided. These spaces provide guidance for the expected length of response.

Please turn over

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SECTION II – 80 marks.

Attempt questions 21-35.

Allow about 2 hour and 25 minutes for this section.

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Question 21 (3 marks)

Two oppositely charged parallel plates separated by 4.00 cm exert a uniform electric field of $1.00 \times 10^4 \text{ N C}^{-1}$.

- (a) What is the voltage across the plates?

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- (b) A charged particle, at rest, is placed between the plates.

The mass of the particle is $9.11 \times 10^{-31} \text{ kg}$ and its charge is $-1.602 \times 10^{-19} \text{ C}$. Assume the force due to gravity on the particle is negligible.

What is the acceleration of the particle?

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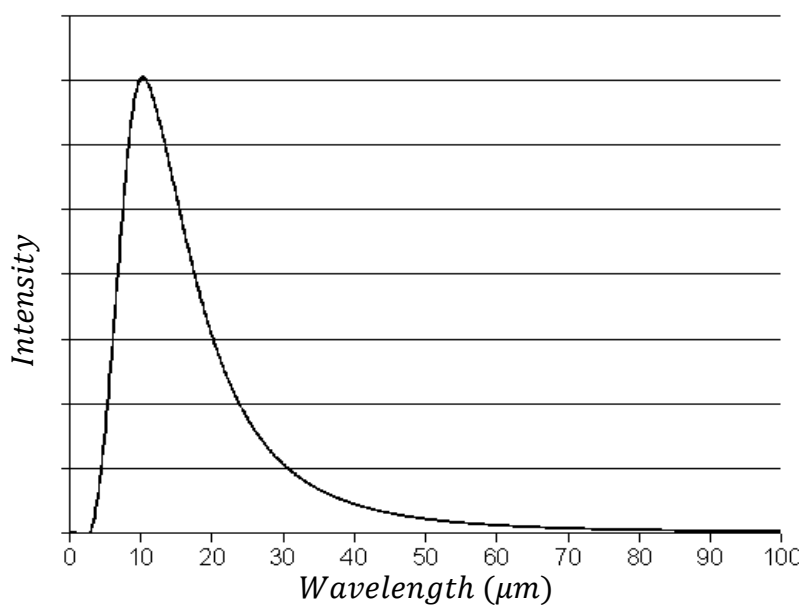
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Question 22 (4 marks)

The diagram below shows the black body radiation distribution of Earth.



Source: <https://commons.wikimedia.org/wiki/File:Blackbody287.pn>

- (a) Show that the temperature of Earth is 290 K

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- (b) A second curve showing the sun's black body radiation is to be drawn on the diagram.

Identify two differences that will be seen between the sun's curve and the Earth's.

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Question 23 (4 marks)

Explain TWO strategies that can improve the efficiency of a transformer.

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Question 24 (5 marks)

Stars undergo many physical and chemical changes as they evolve. For a low mass star, a sequence can be:

Main Sequence → Red Giant → White Dwarf

Explain how nuclear reactions that occur in each phase of the sequence affect the stars' features.

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Question 25 (5 marks)

Many smoke detectors depend on the element Americium-241 to operate. Smoke detectors release steady stream of alpha particles that ionise air molecules as they get impacted.

- a) Americium-241 has a half-life of 432 years. How much americium-241 will remain after exactly 50 years, if a brand new smoke detector contains 2.9×10^{-10} kg americium-241 initially?

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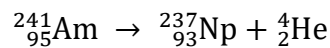
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- (b) Americium-241 undergoes alpha decay into neptunium-237. The following nuclear reaction represents this decay:



The masses of the isotopes in this process are shown in the table:

Isotope	Mass (<i>u</i>)
${}_{95}^{241}\text{Am}$	241.057
${}_{93}^{237}\text{Np}$	237.048
${}_2^4\text{He}$	4.003

The neptunium product created is stationary following the alpha decay.

Determine the speed of the helium atom as it impacts an air particle.

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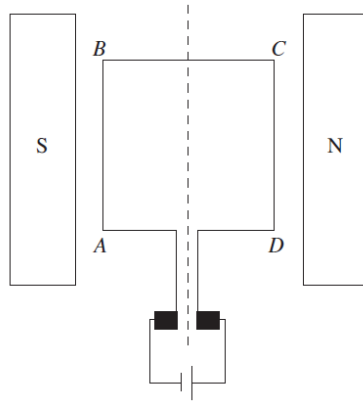
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Question 26 (4 marks)

The diagram shows a simple DC motor with a single loop of wire, $ABCD$.



- a) Compare the forces acting on sides AB and CD in the position shown.

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- b) Outline the effect that doubling the number of loops would have on the maximum torque produced by the motor. Support your answer with a relevant equation.

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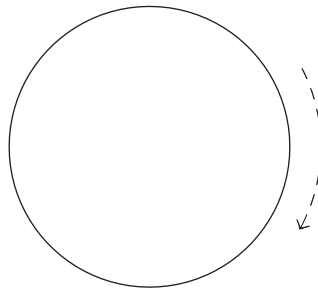
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Question 27 (4 marks)

A horizontal disc with a radius of 0.35 m rotates around its centre at a constant rate of 1.00 revolution per second, as shown in the diagram.



A 0.031 kg object is fixed to the edge of the disc.

- (a) Calculate the magnitude of centripetal force acting on the object.

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- (b) What would happen to the angular speed of the object if it was moved to a point closer to the centre of the disc? Support your answer with a relevant equation.

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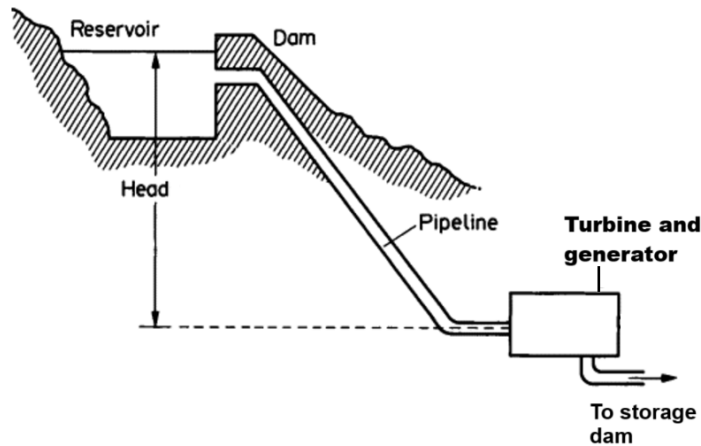
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Question 28 (9 marks)

A farmer on a rural property decides to install a small hydroelectric power system to produce electricity that can be used on the farm. Water from a storage reservoir that is at the top of a hill flows down a pipe, spinning a turbine (or water wheel) that is connected to the armature of a DC generator which produces electrical energy that can be utilised on the farm through the use of an inverter and other control equipment to convert the DC to AC.

At night, when electricity from the grid is cheaper, the system uses the DC generator but runs it as a DC electric motor, to pump water back up to the storage dam for reuse the next day.



A Physics student decided to create a basic model of the farmer's design for a depth study using a raised container of water which would be used to spin a water wheel connected to the shaft of a simple DC generator that had radial permanent magnets. They did not include any additional control systems that the farmer used. Once the container was empty of water, a DC battery was used to run the DC generator as a DC motor to pump the water back up to the container. They did not include any additional control systems that the farmer used.

Using Physics principles and the principle of conservation of energy, explain how the structure of a DC generator produces DC current and why the energy required when the DC generator was run as a DC motor required more energy than the student calculated theoretically.

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24

This image shows a full page of a handwriting practice worksheet. It consists of numerous horizontal dashed lines spaced evenly across the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

This image shows a full page of a document template designed for handwriting practice. It consists of a series of evenly spaced, horizontal black dotted lines running across the entire width of the page. There are no margins, text, or other markings present.



Student Number _____

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Class: Circle one of the following

12Physics25A2 (RP) 12Physics25A3 (JM^cH) 12Physics25A6 (JM^cH)

Merewether High School

2025

Year 12 Physics Trial Examination Section II Answer Booklet – Part B

Instructions

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Please turn over

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Question 29 (3 marks)

Outline Schrödinger's contribution to the quantum mechanical model of the atom.

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Question 30 (5 marks)

With a radius of 476 km and mass of 9.1×10^{20} kg, the dwarf planet Ceres is the largest object in the asteroid belt between Mars and Jupiter.

- (a) If a 9.00×10^2 kg rover landed on the surface of Ceres, determine its gravitational potential energy. **2**

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- (b) Calculate the escape velocity from the surface of Ceres. **3**

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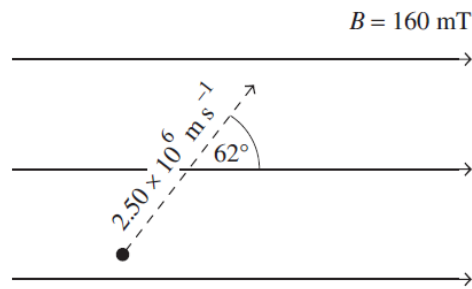
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Question 31 (4 marks)

A charged particle is moving at $2.50 \times 10^6 \text{ m s}^{-1}$ at an angle of 62.0° to the direction of a uniform magnetic field of strength 160 mT, as shown in the diagram.



The particle moves in a helical path in the magnetic field with a radius of 0.286 m.

Calculate the charge-to-mass ratio of the particle.

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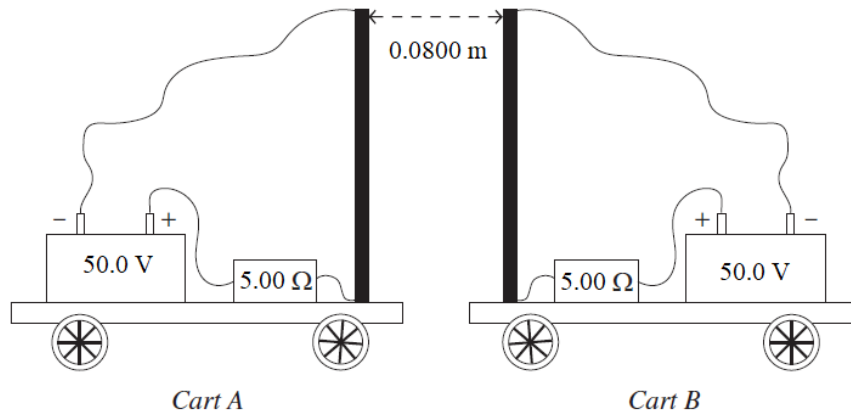
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Question 32 (5 marks)

Two identical carts, *A* and *B*, of mass 0.650 kg each have a fixed 0.500 m long vertical metal rod connected to a 50.0 V battery and $5.00\text{ }\Omega$ resistor.

Initially, the batteries are switched off, and the metal rods are separated by a distance of 0.0800 m , as shown in the diagram.



Determine the initial acceleration of each cart when the batteries are switched on.

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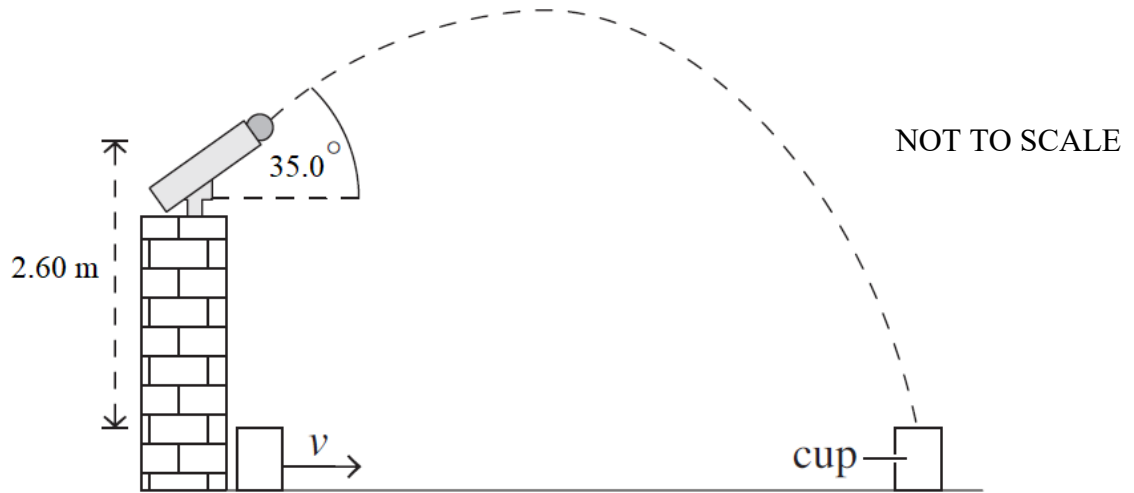
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Question 33 (10 marks)

A spring-loaded ball launcher is fixed at a 35.0° angle on a platform, as shown in the diagram.

When a 46.0 g ball is launched, a target cup is released from the base of the platform and moves along a linear track directly away from the platform at a constant speed. The speed at which the target cup moves along the track can be set by the operator prior to launching the ball.



It is known that the spring in the launcher does 19.4 J of work on the ball.

- a) Show that the launch speed of the ball is 29.0 m s^{-1} if it starts from rest.

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- b) Determine the horizontal speed of the launched ball.

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Question 33 (continued)

c) Calculate the height the ball reaches above the top of the cup. **4**

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d) Determine the distance the target cup must travel so that the ball lands in the cup. **2**

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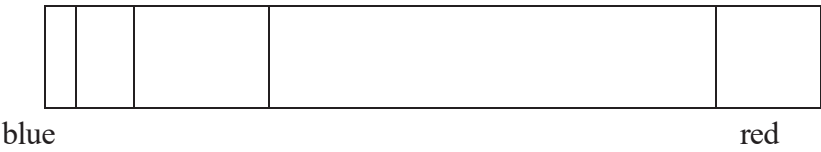
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Question 34 (7 marks)

The hydrogen absorption lines in a star’s spectrum are shown in the diagram.



a) Account for the production of this spectrum.

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(b) Explain how the translational velocity of this star could be determined from its spectrum.

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Question 35 (8 marks)

Analyse how the Geiger–Marsden experiment and Chadwick’s discovery of the neutron provide experimental evidence for the nuclear model of the atom.

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Extra writing space

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Merewether High School

2025

Year 12 Physics Trial Examination Marking Guidelines and Marker's Comments

Multiple-choice Answers

Q. 1	☐ A	☐ B	☐ C	☒ D
Q. 2	☐ A	☒ B	☐ C	☐ D
Q. 3	☒ A	☐ B	☐ C	☐ D
Q. 4	☐ A	☐ B	☐ C	☒ D
Q. 5	☒ A	☐ B	☐ C	☐ D
Q. 6	☐ A	☐ B	☒ C	☐ D
Q. 7	☐ A	☐ B	☒ C	☐ D
Q. 8	☐ A	☐ B	☐ C	☒ D
Q. 9	☐ A	☐ B	☐ C	☒ D
Q. 10	☐ A	☒ B	☐ C	☐ D
Q. 11	☐ A	☐ B	☒ C	☐ D
Q. 12	☒ A	☐ B	☐ C	☐ D
Q. 13	☐ A	☒ B	☐ C	☐ D
Q. 14	☐ A	☐ B	☒ C	☐ D
Q. 15	☐ A	☒ B	☐ C	☐ D
Q. 16	☒ A	☐ B	☐ C	☐ D
Q. 17	☐ A	☐ B	☒ C	☐ D
Q. 18	☐ A	☐ B	☐ C	☒ D
Q. 19	☐ A	☐ B	☒ C	☐ D
Q. 20	☒ A	☐ B	☐ C	☐ D

PART I – 20 marks

Attempt Questions 1-20

Allow about 35 minutes for this part

Use the multiple-choice answer sheet for Questions 1-20

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely using a 2B pencil.

Example:

Question 1

$2 + 4 = ?$ A. 2 B. 6 C. 8 D. 9

Q.1 ☐ A ☐ B ☒ C ☐ D

If you think you have made a mistake, **FULLY ERASE** your incorrect selection, ensuring no marks are left, and then fill in the new answer.

Q.1 ☐ A ☒ B ☐ C ☐ D

-
1. Which of the following evidence suggests protons and neutrons are not fundamental particles?

- A. Protons have a mass of 1.67×10^{-27} kg.
- B. Neutrons have a zero net charge.
- C. Neutrons do not have their own magnetic field.
- D. **Protons and neutrons can undergo beta decay.**

D - Beta decay (conversion between proton/neutron) implied that internal changes have been made and components within protons/neutrons existed

2. A car turning around a horizontal circular bend starts sliding out of control.

Which of the following statements is **incorrect**?

- A. Centripetal acceleration is no longer towards the centre of the curvature of the bend.
- B. **There is no change to the car's centripetal acceleration.**
- C. The coefficient of static friction between the tyres and the road has not changed.
- D. The car has exceeded the maximum threshold velocity for the curvature of the track.

B - If the car begins to slide, it cannot maintain the same centripetal acceleration it had when in control, therefore it is likely to change

3. An ideal transformer has an input voltage of 150 V and an output voltage of 600 V.

What is the ratio of turns in the primary coil to the number in the secondary coil?

A. 1:4

$$\frac{N_p}{N_s} = \frac{150}{600} = \frac{1}{4}$$

B. 1:16

C. 4:1

D. 16:1

4. Which of the following statements best describes a feature of electromagnetic waves according to Maxwell's theory of Electromagnetism?

- A. An oscillating electric field creates a constant magnetic field parallel to the electric field.
- B. An oscillating electric field creates an oscillating magnetic field, but an oscillating magnetic field does not create an electric field.
- C. The found value of the speed of electromagnetic waves was different from the predicted value of the speed of light, so they were distinct phenomena.
- D. An electromagnetic wave's electric field, magnetic field and direction of propagation were all perpendicular to one another.

D - Only D accurately represents the oscillation and propagation of an electromagnetic wave

5. Which of the following best represents the path of electricity from a power station to a home?

- A. Power station → step up transformer → high voltage transmission lines → step down transformer → low voltage transmission lines → home
- B. Power station → step up transformer → low voltage transmission lines → step down transformer → high voltage transmission lines → home
- C. Power station → step down transformer → high voltage transmission lines → step up transformer → low voltage transmission lines → home
- D. Power station → step down transformer → low voltage transmission lines → step up transformer → high voltage transmission lines → home

A - High voltages are used for long-distance electrical energy transmission to reduce the loss of energy. A step-up transformer does this

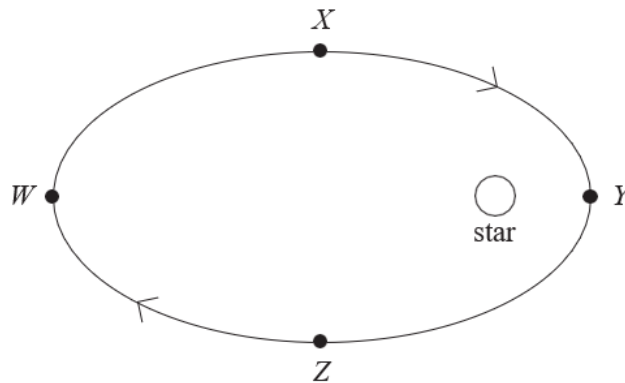
6. An 8500 kg starship drifts towards a planet with a mass of 6.9×10^{27} kilograms and a radius of 72 000 km.

When the starship is 3000 km from the planet's surface, what is its acceleration, assuming the engines are turned off?

- A. 21.8 m s^{-2}
- B. 40 m s^{-2}
- C. 81.8 m s^{-2}
- D. 162 m s^{-2}

C - Since engines are off, the only force felt is the gravitational attraction to the planet. Thus, the net force on the starship is equal to the force of gravity between the starship and the planet. $F_g = GmM/r^2 = ma$ so $a = GM/r^2$ but r = (altitude + radius of planet in metres) which calculates an acceleration of $81.81... \text{ m/s}^2$

7. A comet is in a stable elliptical orbit around a star, as shown in the diagram.

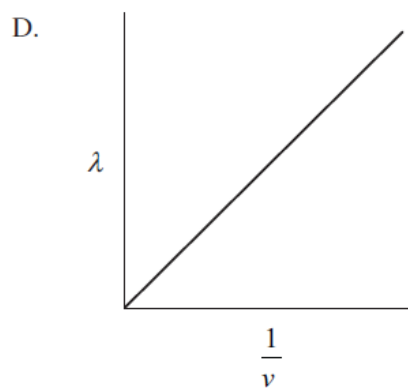
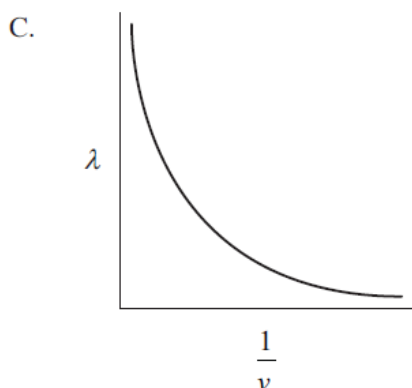
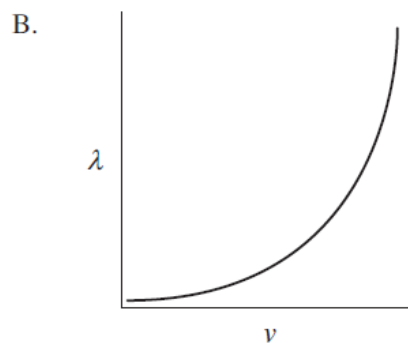
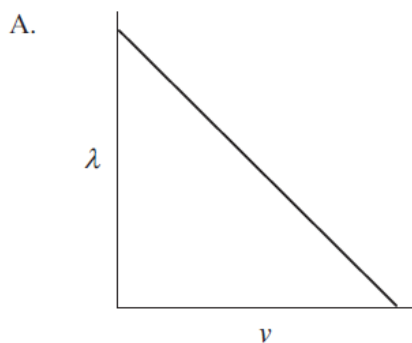


At which position does the comet have the greatest speed?

- A. W
- B. X
- C. Y
- D. Z

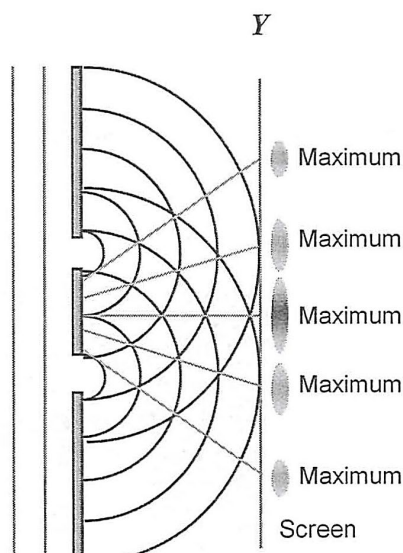
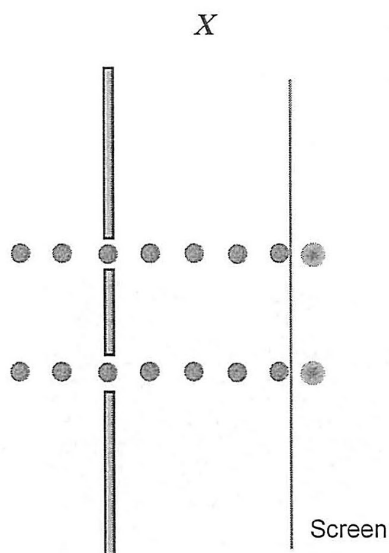
C - An object in a stable orbit has a constant total mechanical energy. As the object in orbit moves closer to the body being orbited, its gravitational potential energy transforms into kinetic energy, and its kinetic energy turns into gravitational potential energy. In accordance with $K = \frac{1}{2} mv^2$, higher kinetic energy corresponds to a higher speed. As position Y is the closest point to the star being orbited, the comet will have the greatest kinetic energy, and thus the greatest speed, at this position.

8. Which graph represents the relationship between the de Broglie wavelength, λ , and speed, v ?



D - According to the de Broglie wavelength ($\lambda = h/mv$) is directly proportional to the reciprocal of speed; that is, $\lambda \propto 1/v$. Plotting λ vs $1/v$ would give a straight line hence graph D

9. Diagrams X and Y below depict possible results of light entering two narrow slits which are very close together.



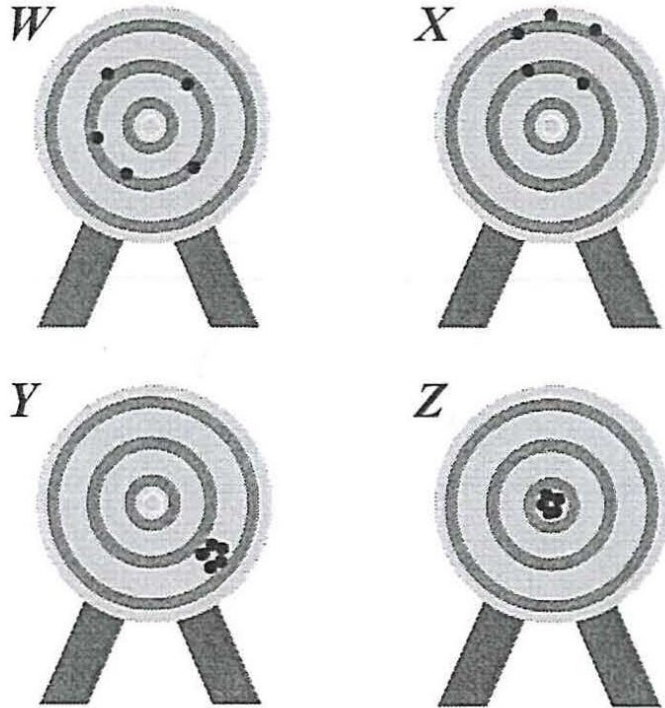
Which statement about these diagrams is correct?

- A. The particle model of light supports both diagrams.
- B. The wave model of light supports both diagrams.
- C. Diagram X shows the results of Young's double slit experiment.
- D. The particle nature of light is not supported by diagram Y.

D – the pattern produced in diagram Y could not be explained by the particle model of light

10.

The diagrams show the results of darts being thrown at a board.

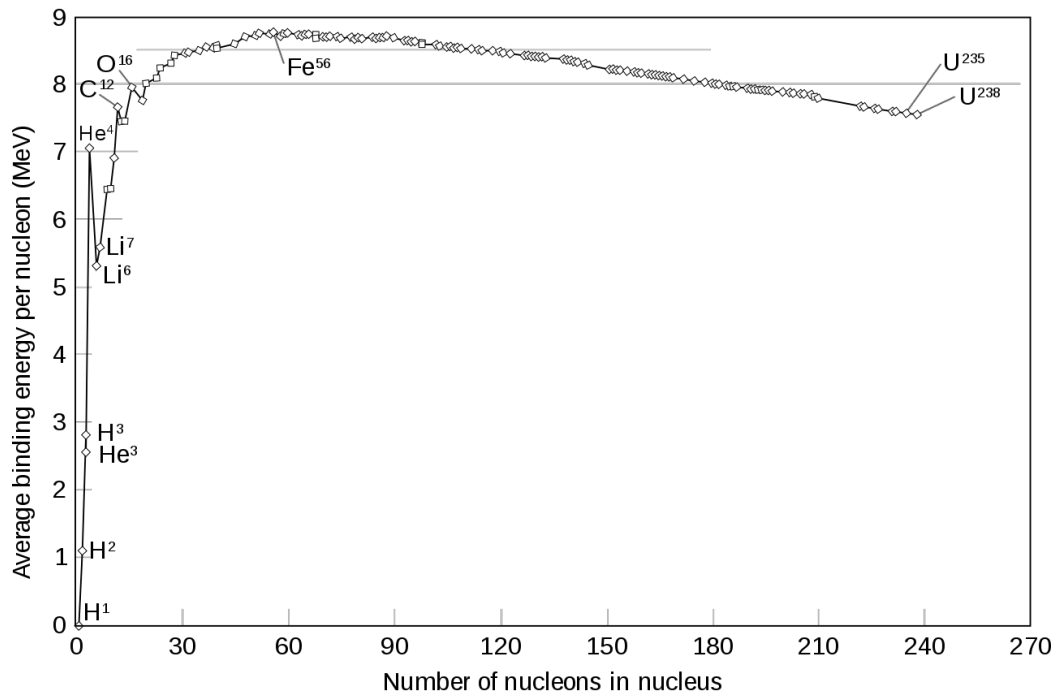


Which statement about these results is correct?

- A. Results W are more reliable than results Y.
- B. Results Y and Z are more reliable than W and X.
- C. Results Z are reliable, while results Y are unreliable.
- D. Results X are less reliable than results W.

Reliable results are highly clumped, so results Y and Z are more reliable than W and X.

11. The diagram shows the average binding energy per nucleon against the number of nucleons inside the nucleus.



Source: https://commons.wikimedia.org/wiki/File:Binding_energy_curve_-_common_isotopes.svg

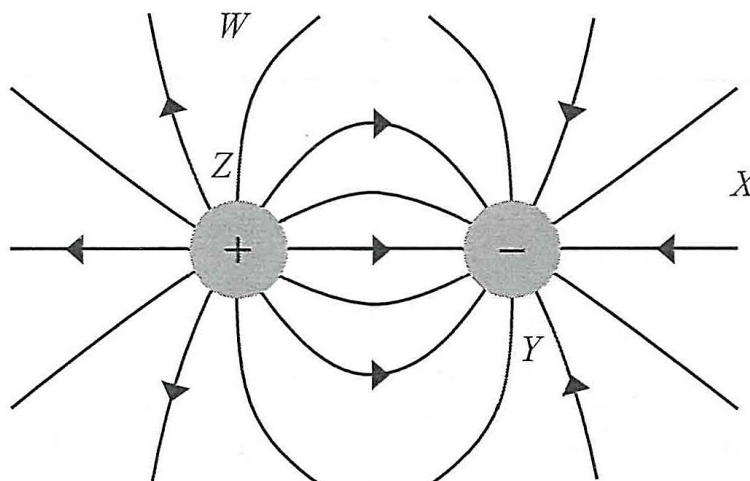
An exothermic reaction releases energy whereas an endothermic reaction absorbs energy from the environment.

Which of the following statements is correct? Had to change option D after the exam as had two answers the same

A.	Nuclear fusion resulting in He^4 is endothermic.	The decay of U^{238} into Th^{234} is endothermic.
B.	Nuclear fusion resulting in He^4 is exothermic.	The decay of U^{238} into Th^{234} is endothermic.
C.	The decay of U^{238} into Th^{234} is exothermic.	Nuclear fusion resulting in He^4 is exothermic.
D.	It cannot be determined whether fusion or fission reactions are endothermic or exothermic from the graphic given.	

C - Only option C correctly states that past iron, exothermic reactions occur during fission, whereas prior to iron, the fusion of Helium-4 is exothermic

12. Four positions are marked within the electric field shown.



A positive charge is moved from one of these positions to another position.

Which move will involve the most work being done by the field?

- A. Moving the charge from *X* to *Y*
- B. Moving the charge from *W* to *Z*
- C. Moving the charge from *Y* to *W*
- D. Moving the charge from *Y* to *Z*.

A – as it asks for work done BY the field and it is a positive charge then the displacement has to be in a direction WITH the field. Moving the positive charge from *X* to *Y* is the only option where the positive charge has work done BY the FIELD. All the other movement are moving AGAINST the field so would have to be done by an external force...

13. Two planets, *A* and *B*, orbit the same star. Planet *A*'s average orbital distance is six times larger than the average orbital distance of planet *B*.

What is the relationship between the orbital periods, *T*, of the planets?

- A. $T_A = 6T_B$
- B. $T_A = 15T_B$
- C. $T_A = 36T_B$
- D. $T_A = 216T_B$

$$\begin{aligned}\frac{r_A^3}{T_A^2} &= \frac{r_B^3}{T_B^2} \\ T_B &= \sqrt{\frac{r_B^3 T_A^2}{r_A^3}} \\ &= \sqrt{\frac{1^3 \times 1^2}{6^3}} \\ &= 0.0680 \\ \frac{1}{0.0680} &= 14.7 \\ \text{Therefore, } T_A &= 15T_B.\end{aligned}$$

14. What is the energy of a gamma ray photon produced from a single positron–electron annihilation?

- A. 0.5 eV
- B. 1.0 eV
- C. 0.5 MeV
- D. 1.0 MeV

$$E = mc^2$$

$$= 2 \times (9.109 \times 10^{-31}) \times (3.00 \times 10^8)^2$$

$$= 1.6396 \times 10^{-13} \text{ J}$$

$$\frac{1.6396 \times 10^{-13}}{1.602 \times 10^{-19}} = 1.0 \times 10^6 \text{ eV}$$

$$= 1.0 \text{ MeV}$$

As two gamma ray photons are produced in a positron-electron annihilation, each photon has 0.5 MeV of energy.

15. A circular coil of wire with a radius of 4.0 cm and 500 circular loops is placed in an electromagnet with the magnetic field decreasing at a rate of 0.200 T s^{-1} . The angle between the plane of the coil and the magnetic field is 30 degrees.

What is the magnitude of the induced emf? (A is the correct answer now but had to be changed after the exam from 0.213 V – was marked as B being correct as that was what had been entered into the marking system – saved changing the marking key)

- A. 0.251 V
- B. 0.435 V
- C. 0.870 V
- D. 1.50 V

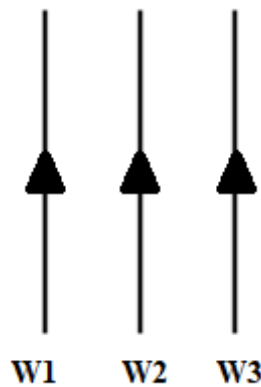
$$\frac{\Delta \Phi}{\Delta t} = \frac{\Delta BA}{\Delta t} \times \cos \theta$$

$$= -0.200 \times \pi \times 0.04^2 \times \cos(90 - 30)$$

$$= -5.027 \times 10^{-4} \text{ Wbs}^{-1}$$

$$\varepsilon = -N \frac{\Delta \Phi_B}{\Delta t} = 0.251 \text{ V}$$

16. Three parallel current carrying conductors result in the middle wire experiencing a net force of 0 N.

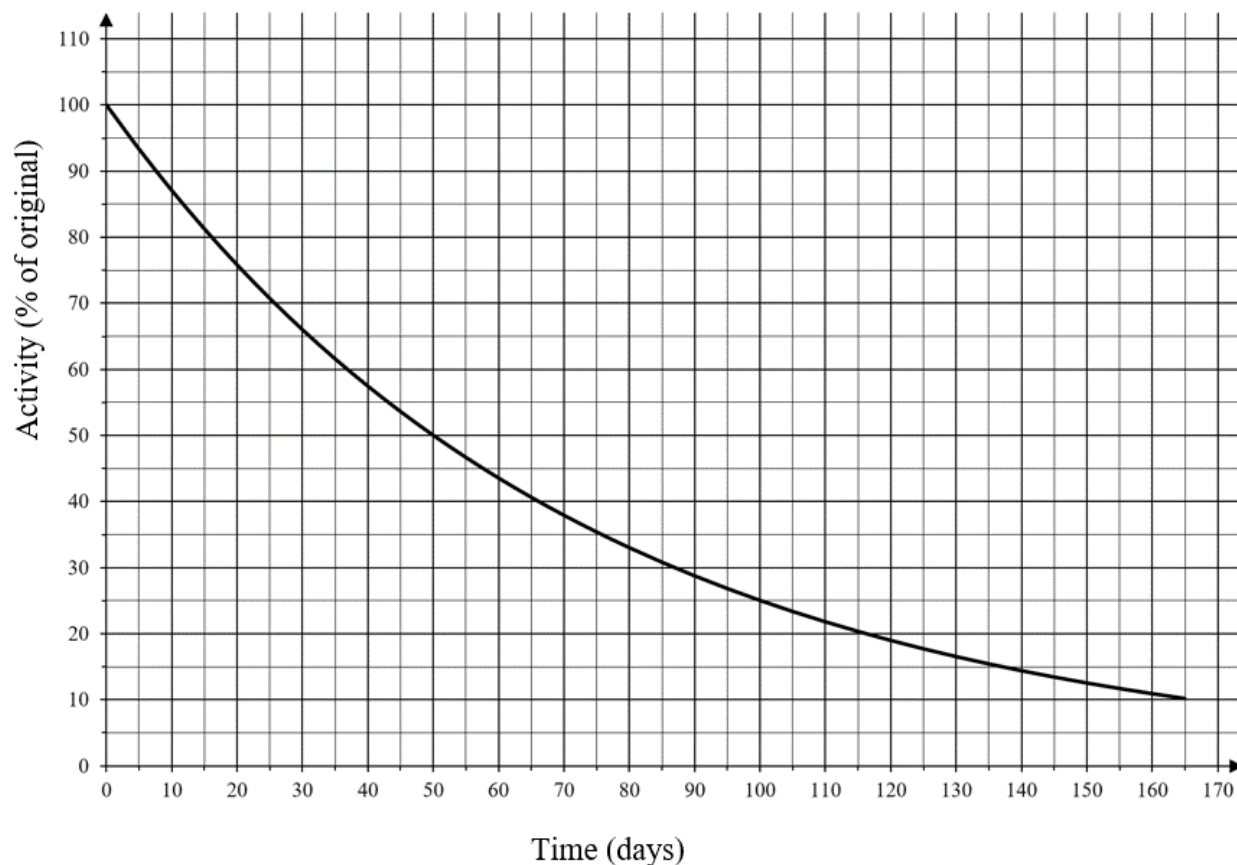


Forces between parallel current carrying conductors – if the current flows in the SAME direction then it is an attraction force and if flowing in opposite directions then it is a repulsive force.

What would be the effect of doubling the current in W_3 AND changing the direction of the current in W_2 ?

- A. W_1 will repel W_2 and W_3 will repel W_2 , resulting in a non-zero net force on W_2 towards W_1 .
- B. W_1 will repel W_2 and W_3 will repel W_2 , resulting in a zero net force on W_2 .
- C. W_1 will repel W_2 and W_3 will attract W_2 , resulting in a non-zero net force on W_2 towards W_3 .
- D. There will be no change to the system.

17. A radioactive sample's activity was observed to decay over time and the results plotted on the graph below.

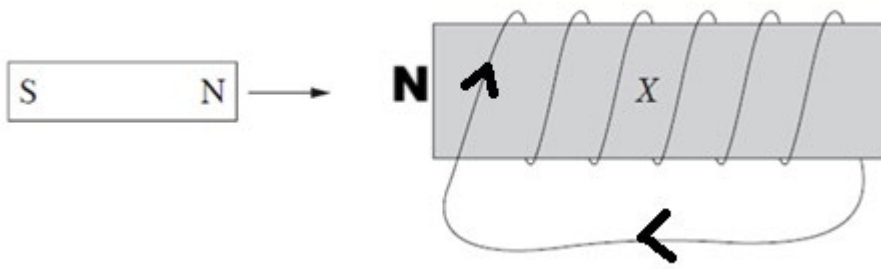


The decay constant, λ , for this radioactive sample is closest to:

- A. $6.0 \times 10^{-3} \text{ s}^{-1}$
- B. $9.6 \times 10^{-6} \text{ s}^{-1}$
- C. $1.6 \times 10^{-7} \text{ s}^{-1}$
- D. $1.6 \times 10^{-7} \text{ day}^{-1}$

Half-life from the graph is 50 days. Using $\lambda = \frac{\ln 2}{t_{1/2}} = 0.0138... \text{ day}^{-1}$. As this is not an answer then we need to convert to units of s^{-1} so $0.0138... / (24 \times 60 \times 60) = 0.00000016045 \text{ s}^{-1}$ so C

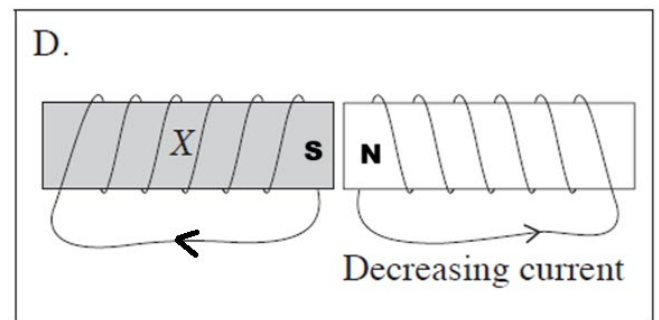
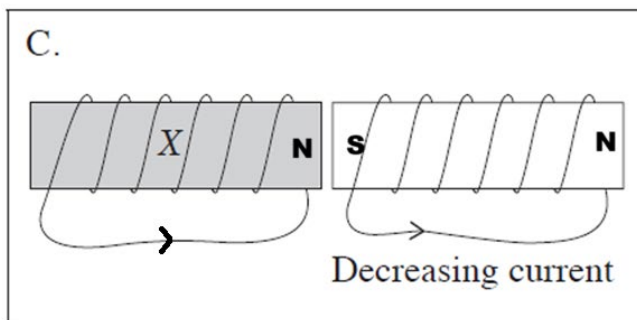
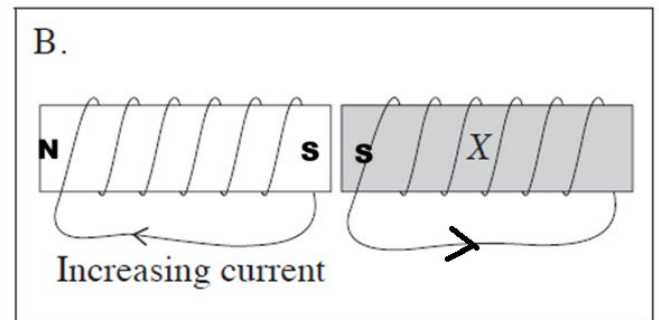
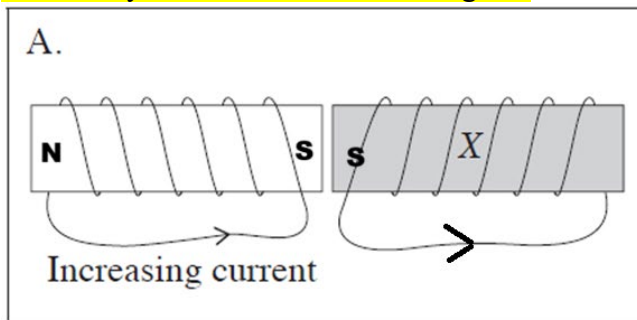
18. The diagram shows a magnet moving towards a coil X.



Moving the magnet to the right in the diagram on the left of this text box would create a North pole at the left end of the solenoid – this results in current flowing as shown.

This action causes a current to be induced in the coil.

Which situation will induce a current in coil X that is in the same direction as the current induced by the movement of the magnet?

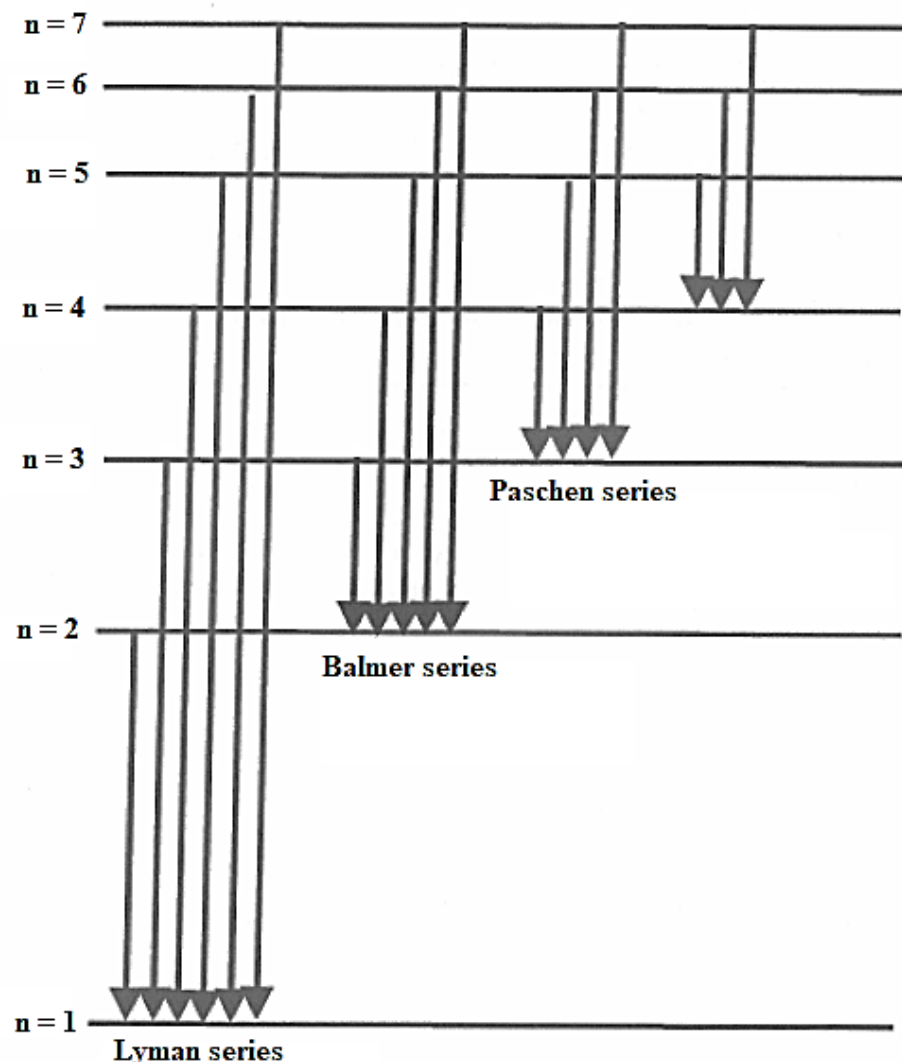


D - When you **increase the current** in the 1st coil then the second coil will induce a current that opposes the increase in flux through the opening of the second coil. In A and B this means that you will end up with a south pole being created in the end of coil X closest to the centre. In C and D, the current in the coil is decreasing so they induced current in coil X has to “replace” the flux lines that it is losing through the opening of coil X. Only D results in the current in coil X flowing in the same direction as the induced current created by the initial magnet movement.

19. In a vacuum chamber there is a uniform electric field and a uniform magnetic field.
- A proton having a velocity, v , enters the chamber. Its velocity remains unchanged as it travels through the chamber.
- A second proton having a velocity, $2v$, in the same direction as the first proton, then enters the chamber at the same point as the first proton.
- In the chamber, the acceleration of the second proton
- A. is zero.
 - B. is constant in magnitude and direction.
 - C. changes in both magnitude and direction.**
 - D. is constant in magnitude, but not direction.

In the first scenario there is no net force on the proton as it stays at the same velocity. If you double the initial velocity, then there will be an unbalanced force, and this would result in circular motion so the velocity would increase and also change direction, so C is the correct answer.

20. The diagram below shows some of the transitions for three series of emission spectra for hydrogen.



An electron in the $n = 6$ energy level of a hydrogen atom relaxed to a lower energy level. In the process, a single photon with energy 3.025 eV was emitted.

Which of the following energy levels did the electron relax to?

A. $n = 2$

B. $n = 3$

C. $n = 4$

D. $n = 5$

$$E_{n_i} = 6$$

$$E = 3.025 \text{ eV}$$

$$3.025 \text{ eV} = E \text{ J}$$

$$1 \text{ eV}$$

$$1.602 \times 10^{-19} \text{ J}$$

$$\therefore E = 4.84605 \times 10^{-19} \text{ J}$$

$$E = hf = h \frac{c}{\lambda} \quad \text{so} \quad \frac{1}{\lambda} = \frac{E}{hc}$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$\therefore \frac{E}{hc} = R \left(\frac{1}{n_f^2} - \frac{1}{6^2} \right)$$

$$\text{so} \quad \frac{4.84605}{6.626 \times 10^{-34} \times 3 \times 10^8} = 1.097 \times 10^7 \left(\frac{1}{n_f^2} - \frac{1}{36} \right)$$

$$0.2222 \dots = \frac{1}{n_f^2} - \frac{1}{36}$$

$$0.2500 \dots = \frac{1}{n_f^2}$$

$$n_f^2 = 3.99$$

$$\therefore n = 2$$

END OF SECTION I

Section II – Part B Marking Guidelines and Marker's Comments – **Marked by Mr McHugh in red pen.**

Question 21(a) (3 marks)

Two oppositely charged parallel plates separated by 4.00 cm exert a uniform electric field of $1.00 \times 10^4 \text{ N C}^{-1}$.

(a) What is the voltage across the plates?

Marker's Comments:

Only several students got this incorrect. Look at sample answer.

Criteria	Mark
Calculates the voltage across the plates	1

Sample answer:

$$\begin{aligned}
 V &= Ed \\
 &= 1.00 \times 10^4 \times 0.04 \\
 &= 400 \text{ V}
 \end{aligned}$$

Question 21(b)

A charged particle, at rest, is placed between the plates.

The mass of the particle is $9.11 \times 10^{-31} \text{ kg}$ and its charge is $-1.602 \times 10^{-19} \text{ C}$. Assume the force due to gravity on the particle is negligible.

What is the acceleration of the particle?

Markers comments:

Students overall did an excellent job on this question. Most of the cohort got full marks and used correct significant figures, units, and direction. Remember MUD (magnitude, units, and direction for a vector). A small percent of students made simple errors e.g. using voltage rather than electric field strength from part (a) of the question.

Criteria	Marks
- Valid formulas to obtain acceleration (1 mark) - Correct units and significant figures (1/2 mark) - Provides a direction as acceleration is a vector (1/2 mark)	2
Provides some relevant information	1

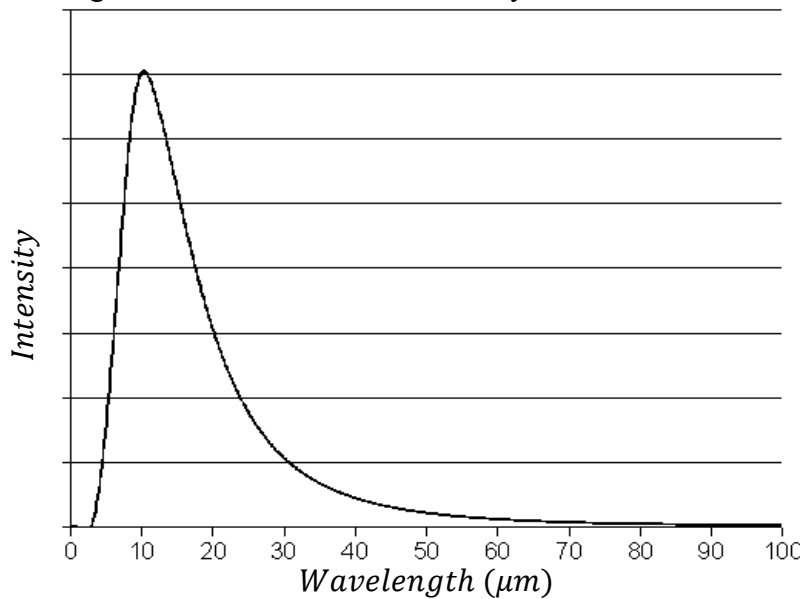
Sample answer:

$$\begin{aligned}
 F &= qE \\
 &= 1.602 \times 10^{-19} \times 1.00 \times 10^4 \\
 &= 1.602 \times 10^{-15} \text{ N} \\
 a &= \frac{F}{m} \\
 &= \frac{1.602 \times 10^{-15}}{9.11 \times 10^{-31}} \\
 &= 1.7585... \times 10^{15} \\
 &= 1.76 \times 10^{15} \text{ m s}^{-2}
 \end{aligned}$$

Therefore, the acceleration of the particle is $1.76 \times 10^{15} \text{ m s}^{-2}$ towards the positive plate.

Question 22 (a)

The diagram below shows the black body radiation distribution of Earth.



Source: <https://commons.wikimedia.org/wiki/File:Blackbody287.png>

(a) Show that the temperature of Earth is 290 K

Marker's Comments:

Only a few students got this incorrect. Check the sample answer below if you had a problem with this question.

Marking Criteria	Marks
• Calculates temperature using Wien's Law	2
• Some relevant information	1

Sample answer:

Using Wien's Law:

$$\lambda = \frac{b}{T}$$

$$T = \frac{b}{\lambda}$$

$$T = \frac{2.898 \times 10^{-3}}{10 \times 10^{-6}}$$

$$T = 290 \text{ K}$$

Question 22 (b)

A second curve showing the sun's black body radiation is to be drawn on the diagram. Identify two differences that will be seen between the sun's curve and the Earth's.

Markers comments:

It was clear the students who had gone back and look at this section of their booklet. (section 7.2 for Module 8 booklet). A lot of students explained the peak would go up but were unable to explain about how the peak would be shifted to the left. See sample answer if you lost marks in this question for an explanation. The peak will increase because the intensity (more photons released) is increasing and the peak shift as going from infra-red into the visible spectrum of EM radiation.

Marking Criteria	Marks
• Explains TWO differences between the black body radiation curves	2
• Provides some relevant information.	1

Sample answer.

- Peak intensity of the sun will be further to the left, since a higher temperature body will release a higher number of photons with higher frequency (lower wavelength). Earth is in the infra-red range while the sun is in the visible range.
- The Peak is not only higher for the sun, but the sun's curve is higher than the Earth's at all wavelengths. Thus, the area under the sun's curve is greater, since a larger, hotter object will emit more photons of all frequencies.

Question 23

Explain TWO strategies that can improve the efficiency of a transformer.

Markers comments:

Overall, this question was done well. It was remarkably similar to a question in the Module 6 MHS booklet that was a previous HSC question. Only a small percentage of the cohort did not get full marks. I strongly suggest reviewing transformers from module 6 if you lost marks on this question and reviewing the sample answer. Think about how to reduce eddy currents when it comes to improving efficiency of transformer. These questions often focus on that area.

Criteria	Marks
• Explains the effect of TWO strategies to improve the efficiency of a transformer	4
• Explains the effect of ONE strategy AND identifies another to improve the efficiency of a transformer	3
• Outlines the effect of strategies to improve the efficiency of a transformer	2
• Provides some relevant information	1

Sample answer:

One method is to reduce energy losses by using a laminated iron core. The thin layers of insulation between the metal laminations interrupt Eddy current loops, making them smaller in size and reduce hysteresis losses, meaning less energy is lost as heat and more is available for transfer to the secondary coil.

Another method is to reduce energy losses by using thicker, or lower resistivity, windings. Thicker wires lower the resistance, so less energy is wasted as heat, which improves the overall efficiency of the transformer.

Question 24 (5 marks)

Stars undergo many physical and chemical changes as they evolve. For a low mass star, a sequence can be:

Main Sequence → Red Giant → White Dwarf

Explain how nuclear reactions that occur in each phase of the sequence affect the stars' features.

Marker comments:

Overall students did this question well; however, it was clear which students had not put in enough time and effort when studying the fusion in stars, HR diagrams and their life cycles. A lot of students failed to understand that a main sequence star that turns into red giant is a lower mass star like our Sun. (Stars greater than about 1.3 solar masses use the CNO cycle). Student also confused the CNO cycle with the triple alpha process.

Common mistakes:

- Not understanding the life cycle of stars on the main sequence. The question is talking about low mass stars.
- Failing to understand star life cycle on a HR diagram. Lots of students need to review star life cycle and HR diagrams.
- Not understanding that stars greater than 1.3 solar masses undergo the CNO cycle which still produces the same amount (very similar) of energy as PP cycle however have a higher mass therefore a great rate of fusion.
- Mixing up the terms fission and fusion.
- Confusing the triple alpha process with CNO cycle. Triple alpha process uses fusion to produce heavier elements such as oxygen, carbon, etc from fusing helium nuclei together after the hydrogen has run out and the star undergoes gravitational collapse.
- The gravitational collapse means that triple alpha process can fuse helium nuclei together which produces more energy but now the star expands as nuclear force is bigger than gravitational force.
- Triple alpha process produces more energy than PP cycle.
- Red giant has lower temperature. Blue is hotter colour than red for stars. A lot of students made this mistake.
- The triple alpha process does produce more energy, but this causes the star to expand leading to a lower energy density meaning the stars colour changes from yellow to red as the surface of the star has reduced.
- The star expands during the triple alpha process and is producing more energy so therefore an increase in luminosity.
- A white dwarf star is the core of red giant after the “shell burning” or nova process of the red giant. This means the core of the star is left over and there is no more fusion taking place. The star cannot fuse heavier elements together. Normally the core is either carbon, oxygen, etc.

Marking Criteria	Marks
Provides an extensive answer explaining how nuclear reactions that stars experience during its life cycle affect the star's features. Answer should include but not be limited to the following points. • Outlines that hydrogen fusion occurs via the proton-proton (PP) cycle in the core of a low mass star. • Explains the balance between nuclear fusion pressure and gravity maintaining stability which causes steady luminosity, temperature, size, and colour or chemical composition of hydrogen and helium. • Explains that after core hydrogen is exhausted, helium fusion begins via the triple-alpha process. Describes core contraction increasing temperature and pressure to start helium fusion. • Explains imbalance between gravity and fusion pressure causes outer layers to expand forming a red giant resulting in changed features: increased luminosity, cooler surface temperature, enlarged size, red colour. • Explains a white dwarf is hot but no longer undergoes fusion due to lack of fusible material resulting in white dwarf's high temperature and low luminosity from residual heat.	5
Provides a comprehensive answer explaining how nuclear reactions that a star experiences during its life cycle affect the star's features but is missing a key piece of information.	4
Provides a sound answer explaining how nuclear reactions that star experience during its life cycle affect the star's features. Has errors in key areas in explain fusion and star features.	3
Provides a limited answer explaining how nuclear reactions that star experience during its life cycle affect the star's features.	2
Provides some relevant information	1

Sample Answer:

During the **main sequence** phase, a low mass star like the Sun fuses hydrogen into helium in its core via the proton-proton (PP) cycle ($4\text{}^1_1\text{H} \rightarrow \text{}^4_2\text{He} + 2\text{}^0_{+1}\text{e} + 2\text{}^0_0\text{v} + 2\text{}^0_0\gamma$). The energy produced from nuclear fusion creates outward pressure that balances the inward pull of gravity. This equilibrium keeps the star stable, with a constant radius, steady luminosity, and a relatively high surface temperature, resulting in a yellow-white appearance.

Once the hydrogen in the core is depleted, nuclear fusion temporarily stops and gravity causes the core to contract. The increased pressure and temperature in the core allow helium fusion to begin via the triple alpha process ($3\text{}^4_2\text{He} \rightarrow \text{}^{12}_6\text{O} + 2\text{}^0_0\gamma$), forming carbon, oxygen etc. The outer layers expand and cool, turning the star into a **red giant**—it becomes much larger and more luminous but its surface temperature drops, giving it a reddish colour. During this phase, the outward pressure from fusion is less than gravity, causing the star's outer envelope to expand.

After the red giant phase, when fusion ceases due to lack of fusible material, the core contracts into a **white dwarf**. A white dwarf is very hot initially (high temperature) but shines only from residual heat, not nuclear fusion, so its luminosity gradually decreases over time as it cools.

Another sample answer

Main sequence stars have relatively consistent size over the duration of this stage. This is due to the balance between inward gravity and outward radiation pressure produced by nucleosynthesis. In a low mass star, the dominant fusion taking place is the proton-proton chain, which is an exothermic

process fusing hydrogen into helium. This fusion creates the energy needed for the star to maintain its size.

Red Giants are large and very luminous. Once the hydrogen concentration decreases, there is less radiation pressure outwards, so gravity begins to collapse the core into a denser, hotter mass. In this phase, due to the increased temperature, some heavier elements begin to fuse one after another in shells. As the star depletes its resources and compacts the core, the outer layers swell and grow. Eventually, the outer layer is ejected, and a white dwarf is formed.

White Dwarfs have a small size, high density, and low luminosity. White dwarfs are the remnant core of a red giant that had insufficient mass to fuse into carbon. There is no fusion taking place. Any luminosity created is from residue heat energy from its temperature.

Question 25(a)

Many smoke detectors depend on the element Americium-241 to operate. Smoke detectors release a steady stream of alpha particles that ionise air molecules as they get impacted.

Americium-241 has a half-life of 432 years. How much americium-241 will remain after exactly 50 years, if a brand new smoke detector contains 2.9×10^{-10} kg americium-241 initially?

Markers comments:

Most of the cohort did this question correctly and clearly knew how to apply this equation. It was a straightforward question. If you made a mistake, check the sample answer. A few students got the significant figures incorrect because they used the value of 50. This is an exact value therefore you can ignore for significant figures. **I did not mark significant figures on this question.**

Criteria	Marks
• Calculates how much Americium will remain after 50 years	2
• Provides some relevant information	1

Sample answer:

$$\lambda = \frac{\ln(2)}{t_{\frac{1}{2}}}$$

$$\lambda = \frac{\ln(2)}{432} = 1.604... \times 10^{-3} \text{ year}^{-1}$$

$$N_t = N_0 \times e^{-\lambda t}$$

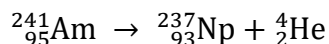
$$N_t = 2.9 \times 10^{-10} \times e^{-1.604... \times 10^{-3} \times 50}$$

$$N_t = 2.676 \times 10^{-10} \text{ kg}$$

Answer to 2 significant figures $N_t = 2.7 \times 10^{-10} \text{ kg}$

Question 25(b)

Americium-241 undergoes alpha decay into neptunium-237. The following nuclear reaction represents this decay:



The masses of the isotopes in this process are shown in the table:

Isotope	Mass (<i>u</i>)
${}_{95}^{241}\text{Am}$	241.057
${}_{93}^{237}\text{Np}$	237.048
${}_2^4\text{He}$	4.003

The neptunium product created is stationary following the alpha decay.

Determine the speed of the helium atom as it impacts an air particle.

Markers comments

Cohort did this question well. A small percentage of students made the common error of not correctly finding the mass defect. If you made this mistake you need to review this process. A lot of students were using the incorrect value from the data sheet or using that information incorrectly. (E.g. 931.5 MeV). Again, a small number of students used the wrong equation to find the velocity of the alpha particle. Review this type of question if you made this mistake. AS it says that the neptunium is stationary after the alpha decay then we can assume that all the energy produced has gone into the kinetic energy of the helium atom.

Criteria	Marks
Calculates the velocity of the helium atom as it impacts an air particle. • Mass defect converted from AMU into kg for the correct value. • Uses $E = mc^2$ to find the total energy. • Uses $\frac{1}{2}mv^2 = K.E = \text{Total energy produced}$ to find velocity of the helium atom.	3
• Provides mostly correct calculations	2
• Some relevant information	1

Sample answer:

$$\text{Mass defect} = 241.057 - (237.048 + 4.003) = 0.006 \text{ u}$$

$$\text{Converting to kg: } 0.006 \times 1.661 \times 10^{-27} = 9.966 \times 10^{-30} \text{ kg}$$

$$\text{Total energy produced} = mc^2 = 9.966 \times 10^{-30} \times (3 \times 10^8)^2 = 8.9694 \times 10^{-13} \text{ J}$$

As Neptunium is considered inert upon creation, the alpha particle must carry this energy produced in the form of kinetic energy, since no other form of energy is created (i.e. gamma).

$$\frac{1}{2}mv^2 = 8.9694 \times 10^{-13} \text{ J}$$

$$v^2 = \frac{2E}{m}$$

$$v = \sqrt{\frac{2E}{m}}$$

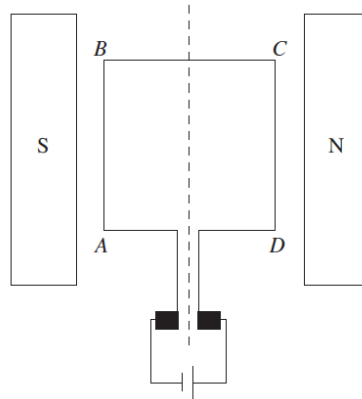
$$v = \sqrt{\frac{2 \times 8.969 \times 10^{-13}}{4.003 \times 1.661 \times 10^{-27}}}$$

$$v = 1.6425... \times 10^7 \text{ m s}^{-1}$$

$$v = 1.643 \times 10^7 \text{ m s}^{-1}$$

Question 26 (a)

The diagram shows a simple DC motor with a single loop of wire, $ABCD$.



- a) Compare the forces acting on sides AB and CD in the position shown.

Marker comments

Most of the cohort did this question correctly. A common problem was students did not consider the need to provide the magnitude of, as well as the direction of, the force in their answer.

Criteria	Marks
• Compares the magnitude and direction of the forces acting on sides AB and CD	2
• Provides some relevant information	1

Sample answer:

The magnitude of the forces acting on sides AB and CD in the position shown are the same. The direction of the force acting on side AB is into the page, whereas the direction of the force acting on side CD is out of the page.

Question 26 (b)

Outline the effect that doubling the number of loops would have on the maximum torque produced by the motor. Support your answer with a relevant equation.

Markers comments

Most of the cohort did this correctly by providing the torque equation of a motor. See sample answer if you made a mistake.

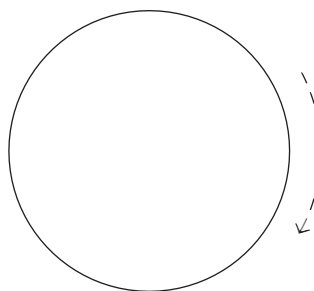
Criteria	Marks
- Torque will double. - Supports answer with relevant equation.	2
Provides some relevant information	1

Sample answer.

In accordance with $T = nIAB$, doubling the number of loops (n) would double the maximum torque (τ) produced by the motor.

Question 27 (a)

A horizontal disc with a radius of 0.35 m rotates around its centre at a constant rate of 1.00 revolution per second, as shown in the diagram.



A 0.031 kg object is fixed to the edge of the disc.

(a) Calculate the magnitude of centripetal force acting on the object.

Markers comments:

Most of the cohort did this correctly. Students who got this question incorrect often made a small mathematical error which resulted in them losing marks. See sample answer. Significant figures were not marked.

Criteria	Marks
- Calculates the centripetal force to 0.43 N. - Correct equations	2
Provides some relevant information	1

Sample answer

$$F_c = \frac{mv^2}{r} \text{ but } v = \frac{2\pi r}{T} \text{ so } F_c = \frac{m(\frac{2\pi r}{T})^2}{r} = \frac{0.031 \times (\frac{2\pi \times 0.35}{1})^2}{0.35} = 0.428... = 0.43 \text{ N}$$

Question 27 (b)

What would happen to the angular speed of the object if it was moved to a point closer to the centre of the disc? Support your answer with a relevant equation.

Markers comments:

About 75% of the cohort got this question correct. However, for 25% of the cohort it was clear that you had not studied this equation. This equation is on your data sheet $\omega = \frac{\Delta\theta}{\Delta t}$ and does not include radius in the formula so therefore changing the radius does not affect the angular velocity. Most students got angular velocity confused with tangential velocity of an object undergoing circular motion. This was a simple question if you knew the correct equation to look at.

Criteria	Marks
• Use correct equation to explain radius does not affect angular velocity. • States that the angular speed is constant at all points on the disc.	2
• Provides some relevant information	1

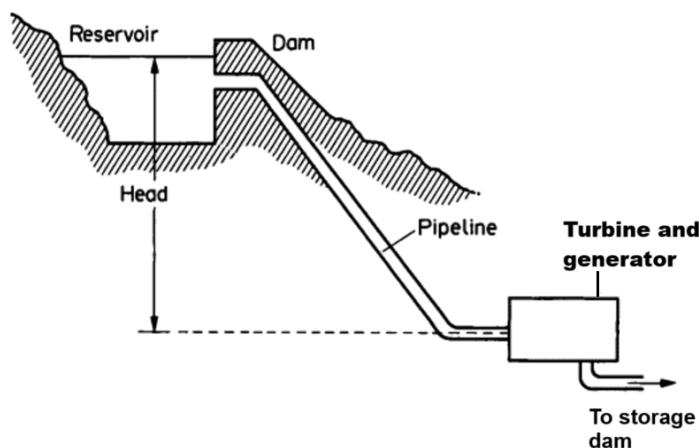
Sample answer

Regardless of where the object is placed on the disc, its angular speed will remain constant. Hence, the object's angular speed will remain at $2\pi \text{ rad s}^{-1}$ if the object was moved to a point closer to the centre of the disc. This is in accordance with ($\omega = \frac{\Delta\theta}{\Delta t}$)

Question 28

A farmer on a rural property decides to install a small hydroelectric power system to produce electricity that can be used on the farm. Water from a storage reservoir that is at the top of a hill flows down a pipe, spinning a turbine (or water wheel) that is connected to the armature of a DC generator which produces electrical energy that can be utilised on the farm through the use of an inverter and other control equipment to convert the DC to AC.

At night, when electricity from the grid is cheaper, the system uses the DC generator but runs it as a DC electric motor, to pump water back up to the storage dam for reuse the next day.



A Physics student decided to create a basic model of the farmer's design for a depth study using a raised container of water which would be used to spin a water wheel connected to the shaft of a simple DC generator that had radial permanent magnets. They did not include any additional control systems that the farmer used. Once the container was empty of water, a DC battery was used to run the DC generator as a DC motor to pump the water back up to the container. They did not include any additional control systems that the farmer used.

Using Physics principles and the principle of conservation of energy, explain how the structure of a DC generator produces DC current and why the energy required when the DC generator was run as a DC motor required more energy than the student calculated theoretically

Markers comments:

- A lot of students found this question difficult.
- Students failed to include key concepts in their answers such as gravitational potential energy of the water or back emf produced when the DC generator is run as a motor.
- Students did not engage enough with the stem of this question and then plan out an answer. A lot of students did not highlight or underline the important parts in the actual question – if you do not answer the question ASKED you cannot gain full marks!
- Students who planned out their answer generally did a lot better in this question. Take the time to plan out a long response question and think about the key points you need to address.
- At the end of writing the question, you can then check that you covered EACH part of the answer that you PLANNED out around the STEM!
- Most of the cohort could explain how a generator worked but found it difficult to link multiple ideas together to provide an extensive answer or compressive answer to gain high marks.

Common mistakes:

- Not explaining the role of gravitation potential energy (GPE) as the source of energy for the system. This was a quite common mistake - over 90% of the cohort did not mention this.

- Outlining the role of the water's kinetic energy (K.E) to spin the turbine therefore the armature of the generator
- Most students could link the structure of a DC generator to producing electrical energy.
- Students did a decent job of explaining electromagnetic induction and how this produced energy. Some of the weaker students fail to clearly outline this point.
- A lot of students again failed to clearly explain how the conservation of energy could be applied to this system that GPE should be converted into electrical energy and visa versa when battery is connected to the system to pump the water back up into the container.
- Not explaining that the motor would have energy losses due to inefficiencies.
- Failing to identify **back emf** in their answer would lower voltage to the motor.
- That back emf would lower the supply voltage to the motor therefore this would lower the current and mean the motor would have to run longer or use more energy to pump the water back up into the holding container.
- Not explaining other energy losses in the system - just saying things like heat is an energy lost. You need to explain the way heat is produced in the system via resistive heating of the wire or via eddy currents in the armature of the motor.
- Need to link energy lost to the motor running for a longer period of time to overcome things such as extra friction, resistive heating of the wires, eddy currents produced, etc.

Marking Criteria	Marks
Provides an extensive answer. Explaining the conservation of energy in this system. How energy transformed to produce a DC current via the structure of DC generator linking the answer to Physics principles. The answer then needs to explain conservation of energy in the DC motor vs the energy produced via the DC generator in terms of energy transformations that make this process inefficient so that more energy is required to pump the water back into the container. Answer should include but not limited to the following points. D.C Generator part: • The energy for the system is provided by the gravitational potential energy of the water which is turned into kinetic when the water is flowing out of the container down the pipe. • The kinetic energy of the water spins the turbine which spins the armature (coil of wire) inside the generator. • D.C generator has the following components a stator which provides the external magnetic field, armature (coil of wire) which can rotate on an axle connected to the water wheel, split ring commutator, brushes which connected to external circuit. • Explain how via electrical energy is produced via Faraday law of induction when a conductor is cutting magnetic flux lines an EMF is produced. • Has equation relevant $\varepsilon = N \frac{\Delta\Phi}{\Delta t}$. • Function of split ring commutator to ensure D.C current in external circuit by changing the polarity. Conservation of energy	9

• Conservation of energy means that gravitational potential energy of the water in the container to produce electrical energy should require the same amount of electrical energy to pump that water back into the container. • However, there are energy losses in the motor that mean the student calculations are incorrect as the motor is not 100% efficient. D.C Motor part: • Explains a valid points about other energy losses in the system such are produced such as resistive heating of the wires, friction, eddy currents being produced in the iron core of the rotor. etc. (Not just heat) • Explains the impact of increase energy lost on pumping the water back up <u>AND/OR</u> • Valid point about back emf lowering voltage at the terminals of the motor therefore increase current to maintain required power of the motor to pump water back up hill. (Or motor having to run for longer period of time) • This led to more current therefore more energy loss as the motor need to be run at higher current or for a longer period this would result in more energy lost due to experiencing frictional and other energy loses for longer.	
Provides a comprehensive answer. Explaining the conservation of energy in this system. How energy transformed to produce a DC current via the structure of DC generator linking the answer to Physics principles. The answer then needs to explain conservation of energy in the DC motor vs the energy produced via the DC generator in terms of energy transformations that make this process inefficient so that more energy is required to pump the water back into the container.	8
Provides a proficient answer. Explaining the conservation of energy in this system. How energy transformed to produce a DC current via the structure of DC generator linking the answer to Physics principles. The answer then needs to explain conservation of energy in the DC motor vs the energy produced via the DC generator in terms of energy transformations that make this process inefficient so that more energy is required to pump the water back into the container.	7
Provides a sound answer. Explaining the conservation of energy in this system. How energy transformed to produce a DC current via the structure of DC generator linking the answer to Physics principles. The answer then needs to explain conservation of energy in the DC motor vs the energy produced via the DC generator in terms of energy transformations that make this process inefficient so that more energy is required to pump the water back into the container.	6
Provides a satisfactory answer. Explaining the conservation of energy in this system. How energy transformed to produce a DC current via the structure of DC generator linking the answer to Physics principles. The answer then needs to explain conservation of energy in the DC motor vs the energy produced via the DC generator in terms of energy	5

transformations that make this process inefficient so that more energy is required to pump the water back into the container. •	
Provides a limited answer. Explaining the conservation of energy in this system. How energy transformed to produce a DC current via the structure of DC generator linking the answer to Physics principles. The answer then needs to explain conservation of energy in the DC motor vs the energy produced via the DC generator in terms of energy transformations that make this process inefficient so that more energy is required to pump the water back into the container. •	4
Provides 3 relevant points from marking criteria	3
Provides 2 relevant points from the marking criteria	2
Provides some relevant information	1

Sample Answer

In the hydroelectric setup, water stored in a raised container has gravitational potential energy. When released, this potential energy converts into kinetic energy as the water flows down the pipe, spinning the turbine connected to the generator. This kinetic energy is mechanical rotational energy which is converted into electrical energy by the generator. Thus, the energy conversion sequence is gravitational potential energy → kinetic energy → mechanical rotational energy → electrical energy.

A DC generator produces direct current because as the armature rotates within the magnetic field created by the radial permanent magnets, an electromotive force (emf) is induced in the armature windings according to Faraday's Law of electromagnetic induction. This happens as there is a change in flux over time as the coils of wire wrapped around the armature cut the magnetic fields lines of the permanent magnets ($\varepsilon = N \frac{\Delta\Phi}{\Delta t}$). The armature's rotation causes the polarity of the induced emf to alternate inside the coil; however, the split-ring commutator reverses the connections to the external circuit at appropriate intervals, ensuring the output current flows in one direction in the external circuit, producing DC current.

The student would have calculated that it should have required the same amount of energy to pump the water back into the container as was produced by the stored gravitational potential energy due to the conservation of energy. (Electrical energy → mechanical rotational energy → kinetic energy of water → gravitational potential energy.). The student's calculations are incorrect as the motor is not 100% efficient.

When the DC generator is run as a motor to pump water back up, more energy is required than the student's theoretical calculation because of several factors such as friction, eddy currents, resistance in the windings causing heat. Eddy currents are produced via Lenz law in the iron core of the motor to oppose motion this produces heat energy, or more resistive heating in the wires as the motor must provide more power against different friction forces in the system. The motor will need to run for longer period to overcome these inefficiencies resulting in greater current draw meaning the motor will require more energy.

AND/OR using back emf as answer.

However, the student may not have considered the effect of the motor producing a back electromotive force (back emf) that opposes the applied voltage; this back emf increases with motor speed and reduces the net voltage across the motor windings ($V_{\text{input}} - \varepsilon = IR$). Back emf in the DC motor is

produced according to Lenz's Law ($\varepsilon = -N \frac{\Delta\phi}{\Delta t}$), which states that an induced magnetic field always opposes the change in flux that caused it. When the motor's armature spins within the magnetic field, it acts like a generator and induces an emf that opposes the applied voltage driving the motor.

This opposing voltage, called back emf, reduces the net voltage across the motor's windings and limits the current ($T=nIAB$ current goes down via these equations). Motor would require a certain amount of power therefore current would need to increase to maintain this power resulting in a greater energy lost. (Also the motor could be run for longer at a lower voltage, with lower current but the motor would need to be run for a longer period resulting in more energy lost from friction, eddy currents, etc)

Section II – Part B Marking Guidelines and Marker's Comments – Marked by Miss Penson in green pen.

Question 29

Outline Schrödinger's contribution to the quantum mechanical model of the atom. (3 marks)

Marker's Comments:

- Quite a few students did not take note of the verb in this question – OUTLINE – this means you need pretty basic facts about what Schrodinger contributed to the quantum mechanical model of the atom. If a question asks about the DEVELOPMENT of the model of the atom, then you need to do “before” and “after” but this didn't really gain you marks in this question as you must answer the question ASKED, not what you think it is ASKING...
- I was quite generous in the marking of this and it was obvious when a student knew what Schrödinger's model was.
- If you are NOT certain who was responsible for an idea then don't mention their name! Some students had Pauli or Schrodinger responsible for the uncertainty principle – this was Heisenberg's work. You could make a statement like “a previous principle stated that the more you knew about an electron's momentum (or velocity) then the less you knew about its position and vice versa” – contradictory information means you cannot get to full marks...
- The quantum model has ORBITALS (3D shapes) and the Bohr model had ORBITS (concentric circular paths) – be careful with these two terms!

Marking Criteria	Marks
• States at least 3 features of the electron cloud model to outline Schrödinger's contribution to the quantum mechanical model of the atom, such as: ○ Names his model as the electron cloud model OR clouds of electrons ○ Mathematical model ○ Gives orbitals where there is a high probability of finding the electrons ○ Electrons as a wave/wave function ○ Different orbitals have different shapes and energy levels. ○ The model incorporates the concept of quantized energy levels, meaning electrons can only exist at specific, discrete energy levels within the atom. ○ treats electrons as both particles and waves (wave-particle duality) ○ incorporates the Heisenberg Uncertainty Principle, which states that it is impossible to know both the exact position and momentum of an electron simultaneously ○ sub energy levels ○ Greater density means a higher probability of finding the electron at that position.	3
• Outlines two features of the quantum mechanical model of the atom	2
• Provides some relevant information or gives one feature	1

Sample answer:

There were many variations accepted to get up to the full 3 marks. Here is one concise example with plenty of features listed.

Schrodinger applied quantum ideas to the pre-existing Bohr-de Broglie model of the atom. Using de Broglie's concept of the electrons orbiting the nucleus as standing matter waves, Schrodinger proposed that the electron would have a probability of existing in a certain 3D region of space (orbital) due to its wave nature, and Heisenberg's uncertainty principle*. This led to the 'electron cloud' model, in which the more dense the 3D 'cloud' region, the more likely the chance of encountering an electron.

* cannot know a particle's position and velocity at the same time

Question 30

With a radius of 476 km and mass of 9.1×10^{20} kg, the dwarf planet Ceres is the largest object in the asteroid belt between Mars and Jupiter.

- (a) If a 9.00×10^2 kg rover landed on the surface of Ceres, determine its gravitational potential energy. (2 marks)

Marker's Comments:

- Gravitational potential energy IS a negative value at ANY point from a large central mass, as by definition, the gravitational potential energy is ZERO at an infinite distance from the centre of the large mass. The only way this definition can be satisfied is if the gravitational potential energy becomes MORE negative as you get closer to the surface!
- A lot of students just "got rid of" the negative sign from the front of the equation for gravitational potential energy for their final answer – YOU CANNOT DO THIS! It is a NEGATIVE value at the surface of the planet!
- Some students didn't round their final answer to two sig figs (due to using 9.1×10^{20} kg in the calculation) OR rounded the calculator answer of 1.1476... incorrectly and gave 1.2 instead of 1.1 as the value before the exponent of $\times 10^8$ – **I went back and re-marked this as there was quite a few other questions that could be marked for the correct number of significant figures. I accepted answers rounded to TWO or THREE significant figures BUT you had to round the calculator answer correctly still!**
- A few students forgot to put UNITS and did not gain full marks.
- A few students wrote N m as the units – which is correct and was marked as correct – but that is an unusual unit (they correctly derived it from the units of G and other quantities in the equation) BUT you would be better off using "J" for joules.

Marking Criteria	Marks
Shows full working (equation, substitution, calc answer, final NEGATIVE answer rounded correctly to 2 or 3 significant figures (when re-marked) and a correct unit) to determine that the gravitational potential energy would be -1.1×10^8 J at the surface of the planet	2
Provides some relevant working	1

Sample answer:

$$\begin{aligned} M &= 9.1 \times 10^{20} \text{ kg} & U &= - \frac{GMm}{r} \\ m &= 9.00 \times 10^2 \text{ kg} & &= - \frac{6.67 \times 10^{-11} \times 9.1 \times 10^{20} \times 9.00 \times 10^2}{476\,000} \\ r &= 476\,000 \text{ m} & &= -11\,476\,235.3 \text{ J} \\ U &= ? & &= -1.1 \times 10^8 \text{ J} \end{aligned}$$

b) Calculate the escape velocity from the surface of Ceres. (3 marks)

Marker's Comments:

- Escape velocity is the minimum speed an object must have to break free from a planet or moon's gravitational field without further propulsion, so I did not mark that you needed a direction – I was very impressed to see that some students were quite particular about this and did include a direction. This shows you are thinking about whether a calculated quantity is a vector or scalar and thinking about fully describing a vector quantity.
- You either had to KNOW the equation is $v_{\text{escape}} = \sqrt{\frac{2GM}{r}}$ or derive it by linking the kinetic energy to the gravitational potential energy i.e. $K = |U|$
- A couple of students mixed up orbital velocity with escape velocity but it is not relevant working so no marks could be awarded – you must answer the question asked to gain marks.
- I also went back and re-marked this part of the question as well in relation to the number of significant figures in the final rounded answer. I accepted answers rounded to TWO or THREE significant figures BUT you still had to round the calculator answer correctly!
- Quite a few students could not correctly round 505 to be 510 when the final answer should be two sig figs as we are using the measured value of 9.1×10^{20} kg which has 2 sig figs in it which will limit our final answer to 2 significant figures...
- You could also use the unrounded value from part a) and use $|U| = |K|$ so $|-1.147... \times 10^8| = \left| \frac{1}{2} \times 9.00 \times 10^2 \times v^2 \right|$ and solve for v to get 505.004... which rounds to 510 m/s.

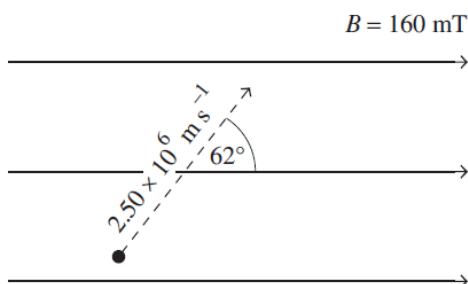
Marking Criteria	Marks
• Show full working to calculate the escape velocity with a correctly rounded number of 2 or 3 significant figures ($5.1 \times 10^2 \text{ m s}^{-1}$ or $5.05 \times 10^2 \text{ m s}^{-1}$).	3
• Calculates the escape velocity with ONE error or omission	2
• Provides some relevant working	1

Sample answer:

$$\begin{aligned}
 K &= |U| \\
 \frac{mv^2}{2} &= \frac{GMm}{r} \\
 v &= \sqrt{\frac{2GM}{r}}
 \end{aligned}
 \qquad
 \begin{aligned}
 M &= 9.1 \times 10^{20} \text{ kg} \\
 r &= 476\,000 \text{ m} \\
 v &= ? \\
 v &= \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 9.1 \times 10^{20}}{476\,000}} \\
 &= 505.00... \text{ m/s} \\
 &= 510 \text{ m/s}
 \end{aligned}$$

Question 31

A charged particle is moving at $2.50 \times 10^6 \text{ m s}^{-1}$ at an angle of 62.0° to the direction of a uniform magnetic field of strength 160 mT, as shown in the diagram.



The particle moves in a helical path in the magnetic field with a radius of 0.286 m.

Calculate the charge-to-mass ratio of the particle. (4 marks)

Marker's Comments:

- **160 mT only has TWO significant figures and it was used in the calculation so the final answer should be TWO significant figures!**
- The centripetal force is supplied by the force on the charged particle by the magnetic field so $F_B = F_c$ and therefore $qvB\sin\theta = \frac{mv^2}{r}$ and this simplifies down to $\frac{q}{m} = \frac{v}{rB\sin\theta}$ where $\frac{q}{m}$ is the charge-to-mass ratio that the questions has asked you to calculate.
- A few students took the component of the VELOCITY but the angle affects how much FORCE is applied to the particle and as we have to equate the centripetal force with the magnetic force. Students who took a portion of the velocity ended up with a value of 4.8 C kg^{-1} rather than 6.2 C kg^{-1} .
- Quite a few students got confused by what the question was asking them to calculate and tried to use a ratio with the ":" sign in the middle.
- Some students did not include UNITS, and you cannot gain full marks without including units...

Marking Criteria	Marks
• Shows full working to calculate the charge-to-mass ratio of the particle as $6.2 \times 10^7 \text{ C kg}^{-1}$ (has TWO sig figures and correct units)	4
• Calculates the charge-to-mass ratio with ONE error or omission in working	3
• Provides some correct steps in a correct procedure	2
• Provides some relevant working	1

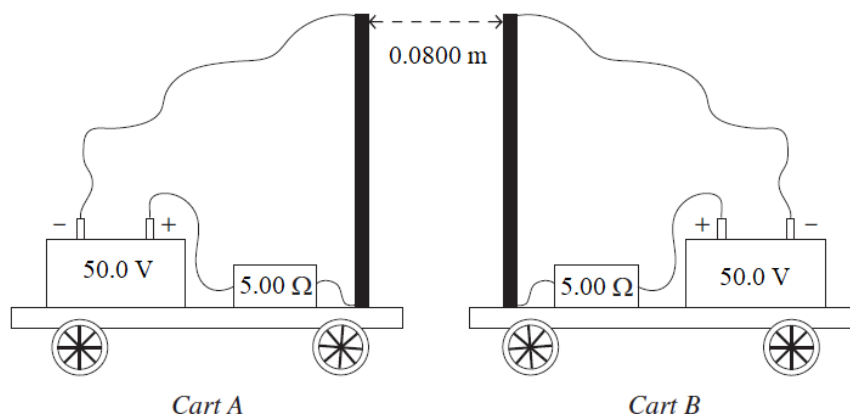
Sample answer:

$$\begin{aligned}
 &\text{Data} \quad F_c = F_B \\
 &v = 2.5 \times 10^6 \text{ @ } 62^\circ \quad \frac{mv^2}{r} = qvB \sin\theta \Rightarrow \frac{mv}{r} = qB \sin\theta \\
 &B = 0.16 \text{ T} \\
 &r = 0.286 \text{ m} \\
 &\frac{v}{r \sin\theta B} = \frac{q}{m} \\
 &\frac{2.5 \times 10^6}{\sin 62^\circ \times 0.16 \times 0.286} = \frac{q}{m} \quad \checkmark \\
 &\frac{q}{m} = 61,875,549.1 \\
 &= 6.2 \times 10^7 \text{ C/kg}
 \end{aligned}$$

Question 32

Two identical carts, *A* and *B*, of mass 0.650 kg each have a fixed 0.500 m long vertical metal rod connected to a 50.0 V battery and 5.00 Ω resistor.

Initially, the batteries are switched off, and the metal rods are separated by a distance of 0.0800 m, as shown in the diagram.



Determine the initial acceleration of each cart when the batteries are switched on. (5 marks)

Marker's Comments:

- Most students knew what they had to do here but made some silly mistakes.
- A few answers that were given did not make any sense whatsoever and this should have been an indication that you had an error in your working somewhere.
- A few students could not write the correct units of m/s^2 or m s^{-2} for the acceleration value calculated and that is just mark leakage.
- Worryingly, a few students thought that they needed to halve the calculated force BUT this is the classic example of Newton's 3rd Law where the force of A on B is equal to the opposite force of B on A i.e. $F_{AB} = -F_{BA}$. The formula calculates the magnitude of the force that acts on A which is also the magnitude of the force acting on B.
- Some students thought that the current running in the same direction resulted in repulsion but that is not the case – it is an attraction force between the two wires.
- A lot of students will have their words of “towards each other” circled as when I was pilot marking to see whether my marking scheme would give a range of marks, I was not sure if that was specific enough when the question asks for the “initial acceleration of EACH cart”. I decided to award this as giving a direction for the accelerations BUT the best answers very specifically said “Cart A has an acceleration of $1.92 \times 10^{-4} \text{ m s}^{-2}$ to the right and Cart B has an acceleration of $1.92 \times 10^{-4} \text{ m s}^{-2}$ to the left”.
- This is a really low acceleration and this is because the force created between parallel current carrying wires IS very small...

Marking Criteria	Marks
Shows full working to determine the magnitude of the initial acceleration to 3 significant figures ($1.92 \times 10^{-4} \text{ m s}^{-2}$) AND the direction of each cart's acceleration (right for Cart A and left for Cart B)	5
- Determines the magnitude of the initial acceleration of each cart OR - has one calculator error in the determination of the magnitude of the initial acceleration to 3 significant figures AND the direction of each cart's acceleration (right for Cart A and left for Cart B)	4
Calculates the correct magnitude of the force acting on each rod	3
Calculates the correct current in each circuit	2
Provides some relevant information	1

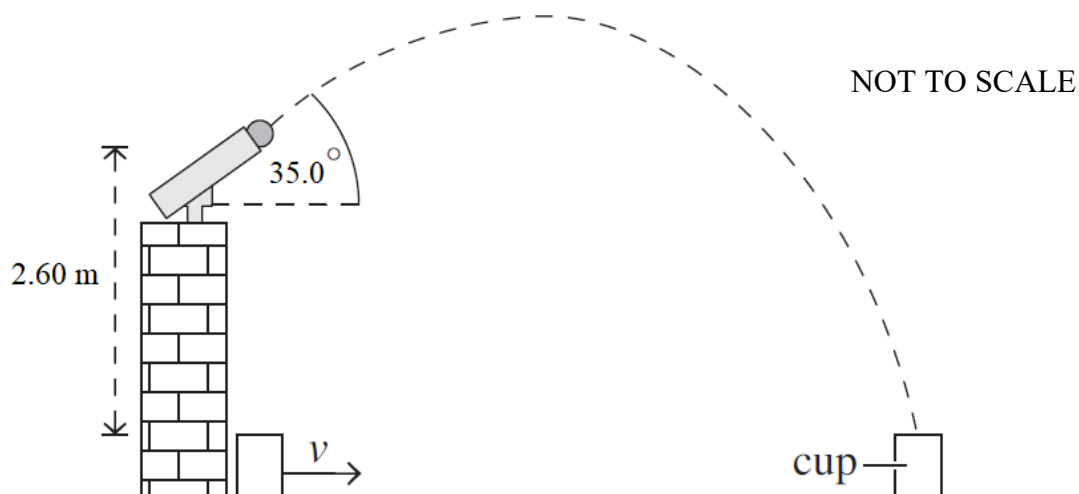
Sample answer:

$$\begin{aligned}
 \text{K12 } I &= \frac{V}{R} = \frac{50}{5} = 10 \text{ A} \checkmark \\
 F &= C \times \frac{\mu_0}{2\pi} \times \frac{I_1 I_2}{r} \\
 &= 0.5 \times \frac{2 \times 10^{-7} \times 10 \times 10}{0.08} \\
 &= 1.25 \times 10^{-4} \text{ N on each cart} \\
 F &= ma \\
 a &= \frac{F}{m} \\
 &= \frac{1.25 \times 10^{-4}}{0.65} \\
 &= 1.92307... \times 10^{-4} \\
 &= 1.92 \times 10^{-4} \text{ m s}^{-2} \text{ (3sf)} \\
 \therefore \text{Cart A will experience an acceleration of } &1.92 \times 10^{-4} \text{ m s}^{-2} \text{ to the right } \checkmark \\
 \text{and Cart B will experience an acceleration of } &1.92 \times 10^{-4} \text{ m s}^{-2} \text{ to the left. } \checkmark
 \end{aligned}$$

Question 33 (10 marks)

A spring-loaded ball launcher is fixed at a 35.0° angle on a platform, as shown in the diagram.

When a 46.0 g ball is launched, a target cup is released from the base of the platform and moves along a linear track directly away from the platform at a constant speed. The speed at which the target cup moves along the track can be set by the operator prior to launching the ball.



It is known that the spring in the launcher does 19.4 J of work on the ball.

- a) Show that the launch speed of the ball is 29.0 m s^{-1} if it starts from rest. (2 marks)

Marker's Comments:

- You must apply the work-energy theorem here, so the work done causes a change in kinetic energy. HSC marking guidelines have emphasised this theorem, so we have been including it in your booklets...
- You should have written $W = \Delta K = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \frac{1}{2}mv^2$ as $u=0$. I accepted $W = \frac{1}{2}mv^2$ as a minimum to gain this mark.

Marking Criteria	Marks
Links work done to a change in kinetic energy	2
Shows full working to prove that the launch velocity rounds to 29.0 m s^{-1}	1

Sample Answer:

$$W = \Delta K = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 \quad \text{so } W = \frac{1}{2}mv^2 \text{ as } u=0$$

$$19.4 = \frac{1}{2} \times 0.0460 \times v^2$$

$$v = 29.042... = 29.0 \text{ m/s}$$

b) Determine the horizontal speed of the launched ball. (2 marks)

Marker's Comments:

- I marked this fairly leniently. You SHOULD have shown a formula to determine the horizontal speed such as $u_h = u \cos \theta$
- Quite a few students did not write their final horizontal speed with 3 significant figures! ALL OF THE MEASURED DATA IN THE QUESTION HAS 3 SIGNIFICANT FIGURES SO ANY ROUNDED, FINAL VALUES GIVEN SHOULD BE 3 SIG FIGS!

Marking Criteria	Marks
• Determines the horizontal speed as 23.8 m s^{-1} with working including showing the calculator answer before rounding to 3 sig figs	2
• Provides some relevant working or has correct working but rounds to a different number of sig figs	1

Sample Answer:

$$\begin{aligned} u_h &= u \cos \theta \\ &= 29.04... \times \cos 35.0 \\ &= 23.79... \\ &= 23.8 \text{ m/s} \end{aligned}$$

Handwritten note: 29.04... m/s at 35.00 degrees

c) Calculate the height the ball reaches above the top of the cup. (4 marks)

Marker's Comments:

- Ideally, you should always carry through unrounded numbers from prior question parts BUT this style of question where they ask you to “show” or “prove” that the launch velocity is 29.0 m/s allows student to continue in the latter parts if they cannot work out the initial launch velocity – it means you can access all the marks instead of being locked out of the question at the very first part!
- As this was the case then I did not penalise anyone who used the rounded value of 29.0 m/s in part b) or c) EVEN if they did calculate it in part a) to a greater number of decimal places than the one decimal place in the given 29.0 m/s.
- You still needed to follow the GUESS technique (Given – Units - Equation – Substitution – Solve) so I needed equation(s), substitution of values shown, calculator answer written, final answer rounded correctly to 3 significant figures and a correct unit to be able to access full marks
- I did not penalise a lack of units for intermediate calculations BUT you should ALWAYS write units for ANY quantity in Physics so some students need to ensure they fix this before their HSC exam!
- Some students rounded a number before the final answer and used it to do a further calculation – we have been very clear that you MUST use the unrounded value in any further calculation!
- A few students did not do the addition of the 2.60 m launch height as an additional, separate step – they simply added it to the line of their final step in the calculation of the height above the launch point and this is MATHEMATICALLY an incorrect statement! Eg got to $19.6 \text{ s} = 277.495\dots$ then calculated $s = 14.157\dots$ but wrote $s = 14.157\dots + 2.60$! Do not do this – I think it was just laziness but this makes it WRONG!
- A few students queried whether the question meant maximum height – the question says “**height** the ball **reaches** above the cup” so this has to be the maximum height...

Marking Criteria	Marks
Shows full working to calculate a height of 16.8 m above the cup (or 16.7 m if used given value of 29.0 m/s to determine the vertical initial velocity) – equation(s), substitution of values shown, calculator answer written, final answer rounded correctly to 3 significant figures and a correct unit for the final height above the cup	4
Has an error or omission in working but has correct process	3
Has the correct process but has several errors	2
Provides some relevant working	1

Sample Answer:

$v = 0 \text{ ms}^{-1}$
 $u = 29.04 \sin 35^\circ \text{ ms}^{-1}$
 $a = -9.8 \text{ ms}^{-2}$
 $s_v = ?$

taking up as positive
 $v^2 = u^2 + 2as$
 $s = \frac{v^2 - u^2}{2a}$
 $= \frac{0^2 - (29.04 \sin 35^\circ)^2}{2 \times -9.8}$
 $= 14.157\dots \text{ m above launch}$

$\therefore h \text{ above cup} = 14.157\dots + 2.60$
 $= 16.75\dots \text{ m}$
 $= 16.8 \text{ m}$

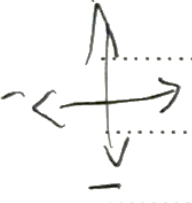
- d) Determine the distance the target cup must travel so that the ball lands in the cup. (2 marks)

Marker's Comments:

- This part should really have been worth 3 marks but it meant that the way it was marked meant that it was very easy to get 1 mark and I marked quite leniently to get up to 2 marks.
- Again, write UNITS for any intermediate variable that you do a calculation for.
- There were a few ways to do this – most calculated time to peak and then time to fall the distance calculated in c) then calculate distance the cup travels or used the quadratic formula to get total time of flight then calculate distance of cup using horizontal velocity and total time.

Marking Criteria	Marks
• Shows most of the working required to calculate the distance the cup has to move (84.3 m)	2
• Provides some relevant working	1

Sample Answer:

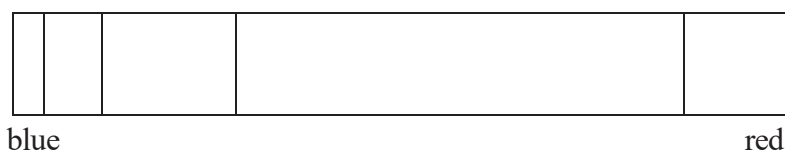
$\times$ ~~to find~~ when ball lands in cup, $s = -2.6\text{ m}$

 $g = -9.8\text{ ms}^{-2}$ in y: $s = ut + \frac{1}{2}at^2$
 $u_y = 29\sin 35\text{ ms}^{-1}$ $-2.6 = u_y t - 4.9t^2$
 $u_x = 29\cos 35\text{ ms}^{-1}$ $4.9t^2 - u_y t - 2.6 = 0$

$$t = \frac{u_y \pm \sqrt{u_y^2 + 4 \times 4.9 \times 2.6}}{2 \times 4.9}$$

$$= 3.5443\dots$$
 ✓
 $s_x = u_x t$
 $= 29\cos 35 \times 3.544\dots$
 $= 84.19\dots$
 $= 84.2\text{ m}$ ✓

Question 34 (7 marks)

The hydrogen **absorption lines** in a star's spectrum are shown in the diagram.



a) Account for the production of this spectrum. (4 marks)

Marker's Comments:

- You must read the question carefully – this is an **ABSORPTION SPECTRUM**.
- Many students accounted for an **EMISSION** spectrum – again, you must answer the question asked – not what you think it is asking!
- You cannot **ACCOUNT** for these black lines in the absorption spectra if you don't link it to the equations related to this phenomenon. We have emphasised that anywhere you can include an equation in an answer you should do so and this is why an equation was needed to get to the full marks.

Marking Criteria	Marks
• ABSORPTION spectrum so energy is absorbed by the star's (cooler) outer layers which contain hydrogen • Hydrogen atoms absorb specific wavelengths of light ($\Delta E = E_f - E_i = \frac{hc}{\lambda}$ or) that correspond to the energy differences between their electron energy levels. • This results in dark lines being created in the star's otherwise continuous spectrum. The result is a spectrum that shows distinct hydrogen absorption lines. • Includes an equation in their account.	4
• Accounts for the production of the ABSORPTION spectrum with ONE error or omission OR • has a full account of how an emission spectrum (rather than the correct absorption spectrum) is produced including an equation	3
• Provides some reasons for the absorption spectrum	2
• Provides some relevant information	1

Sample answer:

Stars produce EM waves across a broad spectrum, including visible light wavelengths (400 nm – 750 nm). As the visible light travels outward from the core of this star, it passes through the star's cooler outer layers, where hydrogen atoms are present. Hydrogen atoms absorb specific wavelengths of light that correspond to the energy differences between their electron energy levels ($\Delta E = E_f - E_i = hf = \frac{hc}{\lambda}$). When photons of specific wavelengths are absorbed by these hydrogen atoms, dark lines are created in the star's otherwise continuous spectrum. Rydberg's mathematical model gives the exact wavelengths for these lines ($\frac{1}{\lambda} = R \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]$). The result is a coloured spectrum that shows distinct black lines for the wavelengths where hydrogen atoms cause energy absorption.

- b) Explain how the translational velocity of this star could be determined from its spectrum. (3 marks)

Marker's Comments:

- I marked this HSC-question style and treated it like a **band 5-6 question**. It was relatively easy to get 2 marks out of the 3 but to get to the full 3 marks you had to **explain** WHAT you would **need to do with the red or blue-shifted lines** to **determine a translational velocity**. This could be stating the effect that **the greater the shift of the spectral lines then the greater the translational velocity** or linking it to the **Doppler effect equation from the data sheet**.
- As it is an **explain** question then you should be asking yourself “what next?” for each step you would undertake until you get to the final step that would allow you to determine the translational velocity of the star – the process is shown below:
 - Measure the wavelengths of the absorption lines from the stars spectrum and compare them to the known values for Hydrogen spectrum lines as measured for a stationary source in a laboratory.
 - If the spectral lines have shifted towards the red end of the spectrum (red shifted), then this means that the star must be moving away from the observer (or if moved towards the blue end of the spectrum (blue-shifted) then the star must be coming towards the observer)
 - The amount that the lines have been shifted can be used to determine the translational velocity of the star ($v = c \frac{\Delta\lambda}{\lambda_0}$) or the Doppler effect formula ($f' = f \frac{(v_{\text{wave}} + v_{\text{observer}})}{(v_{\text{wave}} - v_{\text{source}})}$) can be applied to work out the translational velocity as frequency is linked to the wavelength by $c = f\lambda$

Marking Criteria	Marks
Explains how the translational velocity of the star can be determined from its spectrum by: - Comparing the given spectrum to a hydrogen spectrum created in a Science laboratory (i.e. stationary) or known values for the spectral lines red shift explained i.e. if the 4 lines in the star spectrum have been shifted towards the red end of the spectrum, then the star is traveling away from the observer – must have specifics of WHICH end the lines are moving towards “red-shifted” or “the lines move towards the red end of the spectrum” - The velocity of the star can then be calculated from these shifts in wavelengths. The greater the shift of the spectral lines compared to the known values, the greater the translational velocity OR - links the shift of the observed spectral lines to the Doppler effect equation from the data sheet OR - links it to the equation given in your booklet i.e. $v = c \frac{\Delta\lambda}{\lambda_0}$ Note the best answers also covered blue shift of the spectral lines	3
Shows some understanding of translational velocity and the shifting of absorption lines	2
Provides some relevant information	1

Sample answers:

The Balmer series (the line spectra produced when electrons in a hydrogen atom decrease to shell 2, producing light in the visible spectrum) should be observed and measured in a laboratory. The laboratory spectra should then be compared to the star's spectra. Taking the lab spectra as the wavelengths in the lab spectra as the 'true' value, the degree of red doppler shift in the star's spectra can be determined. If the ~~star~~^{spectra} is red-shifted, the star will have a translational velocity moving away from Earth. The exact velocity can be determined with Hubble's Law, which states that a greater degree of red shift will ~~increase~~ correspond to an increased velocity. If the ~~star~~^{spectra} is blue shifted, then the star has a translational velocity moving towards the Earth.

* Red-shifted: all lines in the ^{star's} spectra have a larger wavelength compared to lab spectra

* Blue-shifted: all lines in the star's spectra have a smaller wavelength compared to lab spectra.

If a star has a translational velocity, it will either be moving towards, or away from the observer. You can determine if a star is moving ^{or towards you} away by using the wavelengths in its absorption spectrum and comparing them to known values of the wavelengths of the spectra of elements present ~~in~~ⁱⁿ the star. If all the wavelengths are longer/higher than the known values, then the spectrum is red-shifted and is moving away from the observer and if all the wavelengths are shorter, then the spectrum is blue shifted and the star is moving towards the observer. This works due to the doppler effect by finding the frequencies of the known wavelengths ($f = \frac{c}{\lambda}$) and that of the observed wavelengths ~~and~~ from the spectrum and hence finding the velocity of the star using $f' = f \frac{(v_{\text{wave}} + v_{\text{observer}})}{(v_{\text{wave}} \pm v_{\text{source}})}$.

This spectrum can be compared to the normal absorption spectrum of hydrogen to determine translational velocity. ~~The more red~~ spectral lines from receding stars are red shifted proportional to recession velocity while lines from approaching stars are more blue shifted. By determining whether this spectrum is red or blue shifted, its direction can thus be determined. Its recession velocity can be determined using $\frac{\Delta \lambda}{\lambda} \times c = v$ where $\Delta \lambda$ is shift in wavelength.

Question 35 (8 marks)

Analyse how the Geiger–Marsden experiment and Chadwick’s discovery of the neutron provide experimental evidence for the nuclear model of the atom.

Marker’s Comments:

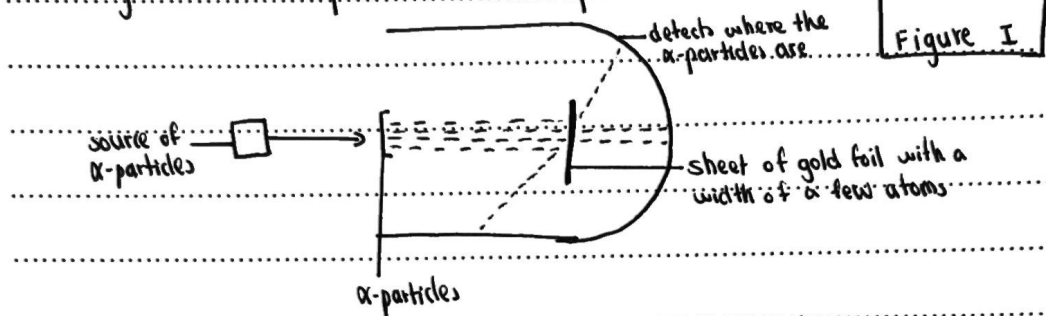
- You cannot bluff a Physics marker – you must get the correct components of experiments named, state the results and what was interpreted from these results! You must know specifics!
- Analyse means “Identify components and the relationship between them; draw out and relate implications” – to answer this question completely you needed to tell me **WHY these experiments made scientists move to the nuclear model over the currently accepted Thomson “plum pudding model”**. I was looking for some mention of what the previous model was and indicated that I had found it by writing “NM” and a tick near the information.
- I broke the question up into 3 parts – Geiger-Marsden experiment (GM), Chadwick’s experiment to discover the neutron (C) and the need for a new model (NM). This way you could match up the following marking scheme with the ticks and crosses on your page to get your final mark out of 8.
- Analyse is one of the highest order verbs – you need to explain the experiments first and then draw out the implication of the results of these experiments and why they supported the nuclear model.
- Diagrams make long responses so much easier to mark – USE THEM!
- A few students mixed up the Davisson & Germer experiment which could not gain marks as it is not relevant to these experiments.
- In some points in the marking scheme I was looking for two points of information – such as, it is vital to know that the interpretation of the observations from the GM experiment meant that Rutherford’s model of the atom was “a **very, small, dense, positive nucleus that contained most of the atom’s mass**, had electrons orbiting in concentric, circular orbits and that **the atom was mostly empty space**”

Marking Criteria	Marks
Geiger–Marsden experiment: (GM) Experiment – fired positive alpha particles at very thin gold foil (and used a detector to determine the path they took) Results – most alpha particles travelled straight through ; some deflected through small angles , and some returned in the direction they were fired from Interpretations of results – atoms must have a very small, dense, positive nucleus to have the repulsion force to repel the +ve alpha particle and be mostly empty space as most alpha particles went straight through.	3
Chadwick’s discovery of the neutron: (C) Experiment – fired alpha particles at beryllium , Results – high-energy “unknown radiation” was produced (accepted unknown particle as well) which ejected protons from paraffin Interpretations of results – used laws of conservation of energy and momentum to determine that the “unknown radiation” were in fact Neutral particles similar in mass to the proton which are what we now call the neutron	4
Hence a need to move to the nuclear model of the atom (NM - new model needed) Describes how Thomson’s plum pudding model (accepted plum pudding model or Thomson’s model or a description of positive matrix with embedded negative electrons) had to be changed to explain the experimental evidence from the two experiments above and move to the nuclear model.	1

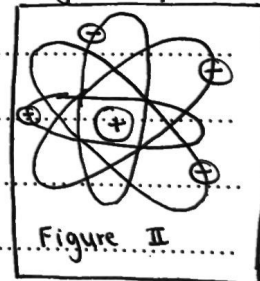
Sample Answer:

Question 35 (continued)

Prior to these experiments, the accepted model of the atom was the plum pudding model, which described the atom as a positively charged matrix, with negatively charged atoms dispersed throughout. This is shown in Figure I. The Geiger-Marsden experiment was set up as follows.



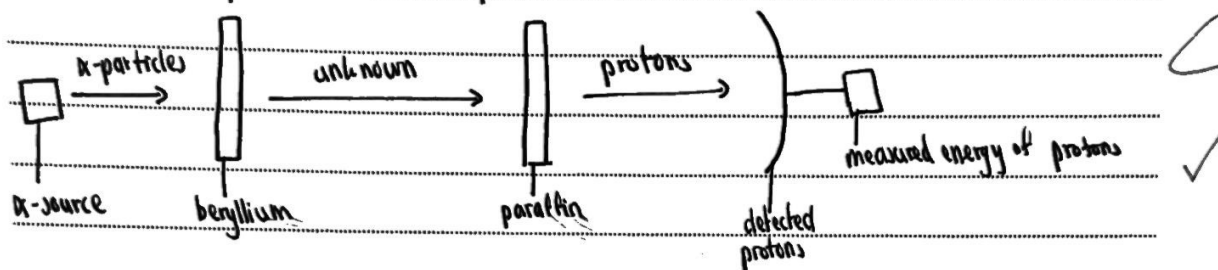
The experimental results were expected to show that all α particles passed through the gold foil undeflected. However, the actual results showed that while most particles passed straight through the gold foil, some were deflected and some were even reflected. Rutherford analysed these results and proposed a new model of the atom. He suggested that the atom was mostly composed of empty space, which is why most α -particles passed through undeflected. However, the positive charge of the atom was not spread out in a matrix but instead localised in a small, dense, positively charged nucleus. If α -particles hit



to this nucleus they were repelled, causing their deflection or reflection. Rutherford thus proposed the first nuclear model of the atom (Figure II) to show the atom as being composed of a positive nucleus with electrons orbiting around it.

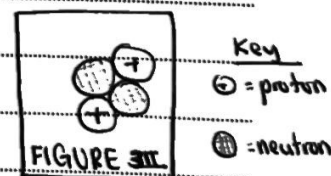
35 continued

However, a problem with this model was that a nucleus only composed of protons should be highly unstable and ~~repel each other~~ as each proton repels each other. Chadwick's experimental set up was as follows.



Initially, the unknown radiation was hypothesised to be gamma rays. This is because the radiation was not deflected by electric or magnetic fields, so must have been neutral. However, Chadwick realised that the protons ejected from the paraffin had an energy of 5 MeV. For this to be possible, the gamma ray would have needed an energy of 50 MeV. But the initial α-particles only had 5 MeV, so the gamma rays could not have more energy otherwise the Law of Conservation of Energy would not hold. ∴ (Unknown could not be γ rays) Chadwick used the Law of Conservation of Momentum to determine that the unknown radiation was a particle, with a mass slightly greater than a proton (0.002×10^{-27} kg greater). This unknown particle was named a neutron. This discovery contributed to the nuclear model.

as the nucleus was described as being composed of protons and neutrons, as shown in Figure III. This resolved why the nucleus was stable, as the neutrons increased strong force and decreased electromagnetic repulsion.



Therefore the nuclear model of the atom now consists of a small, dense, positively charged nucleus ~~with~~ composed of protons and neutrons, surrounded by orbiting electrons. This agreed with the evidence provided by the Rutherford-Mendenhall experiment and Chadwick's experiment.